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Turmoil before the boring billion: Paleomagnetism of the 1880–1860 Ma Uatumã event in the Amazonian craton

Paul Y.J. Antonio [a,b,*], Manoel S. D'Agrella-Filho [a,**], Ricardo I.F. Trindade [a], Anne Nédélec [b], Davis C. de Oliveira [c], Fernando F. da Silva [c], Matteo Roverato [d], Cristiano Lana [e]

[a] Instituto de Astronomia, Geofísica e Ciências Atmosféricas (IAG), Universidade de São Paulo (USP), Rua do Matão, 1226, 05508-090 São Paulo, SP, Brazil
[b] Géosciences Environnement Toulouse (GET), Université de Toulouse, UPS, CNRS, IRD, CNES, 14 Avenue E. Belin, 31400 Toulouse, France
[c] Instituto de Geociências (IG), Universidade Federal do Pará (UFPA), CEP-66075-900 Belém, Pará, Brazil
[d] Instituto de Geociências (IGc), Universidade de São Paulo, Rua do Lago, 562 Cidade Universitária, 05508-080 São Paulo, SP, Brazil
[e] Departamento de Geologia, Universidade Federal de Ouro Preto (UFOP), Morro do Cruzeiro, Ouro Preto 35400-000, MG, Brazil

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ABSTRACT

The Uatumã silicic large igneous province (SLIP) has covered about 1,500,000 km^2 of the Amazonian craton at *ca.* 1880 Ma, when the Columbia/Nuna supercontinent has been assembled. Paleomagnetic and geochronological data for this unit were obtained for the Santa Rosa and Sobreiro Formations in the Carajás Province, southwestern Amazonian craton (Central-Brazil Shield). AF and thermal demagnetizations revealed northern (southern) directions with high upward (downward) inclinations (component SF1), which passes a 'B' reversal test, and is carried by magnetite and SD hematite with high-blocking temperature. This component is present on well-dated 1877.4 ± 4.3 Ma (U-Pb zrn - LA-ICPMS) rhyolitic lava flows, providing the SF1 key paleomagnetic pole (Q = 6) located at 319.7°E, 24.7°S (A_{95} = 16.9°). A second southwestern (northeastern) direction with low inclination (Component SF2) was obtained for a well-dated 1853.7 ± 6.2 Ma (U-Pb zrn - LA-ICPMS) dike of the Velho Guilherme Suite. This component also appears as a secondary component in the host rhyolites of the Santa Rosa Fm and andesites of the Sobreiro Fm at the margins of the dike previously dated. Its primary origin is confirmed by a positive baked contact test, where a Velho Guilherme dike crosscuts the 1880 Ma andesite from the Sobreiro Formation. The corresponding SF2 key pole is located at 220.1°E, 31.1°S (A_{95} = 5°) and is classified with a reliability criterion Q = 7. The large angular distance between the almost coeval (difference of ~25 Ma) SF1 and SF2 poles implies high plate velocities (~39.3 cm/yr) which are not consistent with modern plate tectonics. The similar significant discrepancy of paleomagnetic poles with ages between 1880 and 1860 Ma observed in several cratons could be explained by a true polar wander (TPW) event. This event is the consequence of the reorganization of the whole mantle convection, and is supported by paleomagnetic reconstructions at 1880 Ma and 1860 Ma and also by geological/geochronological evidence.

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1. Introduction

Understanding the position of continents through time is crucial to constrain the geodynamics of the Earth. Supercontinents assembly has punctuated Earth (Nance et al., 2014) with at least three supercontinents being recognized: Pangea (*ca.* 300–200 Ma), Rodinia (*ca.* 1000–700 Ma) and Columbia/Nuna (*ca.* 1800–1400 Ma). The existence of supercratons during the Archean (*ca.* 2700 Ma) is also speculated (Evans et al., 2016a; Meert, 2012, 2014). While the configurations of Pangea and Rodinia are the object of a relative consensus (Domeier et al., 2012; Li et al., 2013), the timing and the exact configuration of Columbia/Nuna (hereafter called Columbia following Meert (2012)) is still debated. The main geological argument for the existence of this supercontinent is the presence of 2100–1800 Ma orogens in most continents (Zhao et al., 2002; Zhao et al., 2004). No changes in ocean and atmosphere composition were observed during the Columbia supercontinent and even more longer (1800–800 Ma) which is referred as the "Boring billion" (Holland, 2006). This period is tectonically far from boring and to precisely constrain its paleogeography a more comprehensive paleomagnetic database (such as the PALEOMAGIA) for the Paleo and Mesoproterozoic covering most continental blocks worldwide is needed (Veikkolainen et al., 2014b). Presently, few key poles are available and they are unevenly distributed across the globe. Most

* Correspondence to: P.Y. J Antonio, Instituto de Astronomia, Geofísica e Ciências Atmosféricas (IAG), Universidade de São Paulo (USP), Rua do Matão, 1226, 05508-090 São Paulo, SP, Brazil.
** Corresponding author.
E-mail addresses: paulantonio0931@gmail.com (P.Y.J. Antonio), dagrella@iag.usp.br (M.S. D'Agrella-Filho).

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models therefore use pairs of coeval poles from cratonic blocks to test reconstructions, but this implies that the models are not unique with many alternatives (For example, see models proposed by Pisarevsky et al. (2014) and Bispo-Santos et al. (2014b) for the Amazonian craton at *ca.* 1790 Ma). Given these uncertainties, the time of maximum packing of Columbia is not a consensus, with some authors proposing it occurred at *ca.* 1780 Ma (Bispo-Santos et al., 2014b; Zhang et al., 2012), while others think it was reached only at around 1580 Ma (Pehrsson et al., 2016; Pisarevsky et al., 2014). The detrital zircon database, however, favors the first hypothesis (Condie and Aster, 2010; Hawkesworth et al., 2010).

The Laurentia-Baltica connection forms the core of Columbia. The extensive paleomagnetic database for these cratons supports a long connection between them, from 1780 to 1260 Ma (Buchan et al., 2000; Pisarevsky and Bylund, 2010; Salminen et al., 2014; Salminen et al., 2015). Recently, paleomagnetic studies on rock units from the Amazonian craton suggest that this cratonic block was also part of Columbia (Bispo-Santos et al., 2008; Bispo-Santos et al., 2012; Bispo-Santos et al., 2014a; Bispo-Santos et al., 2014b; D'Agrella-Filho et al., 2012; D'Agrella-Filho et al., 2015; D'Agrella-Filho et al., 2016a, 2016b). Several lines of evidence suggest a connection between the Amazonian craton and Baltica in a SAMBA (South-America-Baltica) configuration (Johansson, 2009, 2014), which was positively tested by the 1790 Ma Avanavero pole (Bispo-Santos et al., 2014b). Paleomagnetic and geological evidences for alternative models to SAMBA exists (Bispo-Santos et al., 2008; Bogdanova et al., 2015; Pisarevsky et al., 2014). The Amazonian craton is also usually linked with West Africa, along a connection between the Guri (Guiana Shield) and Sassandra (West Africa) shear zones as updated by Bispo-Santos et al. (2014a) using the new 1960 Ma Surumu pole.

If uncertainties remain in the Columbia configuration, the paleogeography prior to this supercontinent assembly is still more uncertain given the independent drift paths of most continental blocks (*e.g.* Laurentia, Baltica, Amazonian, West Africa). In such case, match of large igneous provinces (LIPs) can be used as an additional tool to correlate continental blocks, using similarities in their geochronology and geochemistry ("LIP barcode"). On the other hand, the orientation of dike swarms can be applied, as they generally herald the breakup of large continental masses (Ernst et al., 2013; Söderlund et al., 2016). For instance, by ~1880 Ma a large intraplate and plate margin magmatism occurs in numerous Archean cratons worldwide that provides paleomagnetic data that show significant angular distances between poles in the same cratonic unit with only slightly different ages. The best-known example is the Coronation loop for the Slave craton (McGlynn and Irving, 1978; Mitchell et al., 2010), but similarly anomalous results are also noted for Kalahari (Hanson et al., 2004; Hanson et al., 2011) and Superior cratons (Halls and Heaman, 2000; Hamilton et al., 2009), suggesting a true polar wander (TPW) signal (Mitchell et al., 2010). These anomalous apparent polar wander paths (APWPs) have been interpreted as either (i) rapid plate motions, much faster than today, (ii) TPW events when the whole solid Earth (mantle and crust) rotates in relation to the liquid outer core to accommodate a change in global moment of inertia (Mitchell et al., 2010), or (iii) the absence of a stable GAD (geocentric axial dipole) field ("multipolar field"), including the possibility of an equatorial dipole or "hyper" frequent polarity reversals (Abrajevitch and Van der Voo, 2010; Bazhenov et al., 2016; Driscoll, 2016; Halls et al., 2015).

Here we focus on the paleomagnetic record of the Amazonian craton for the *ca.* 1880–1860 Ma interval, when a large intraplate magmatism has covered most of its surface defining the Uatumã silicic (~felsic) large igneous province (SLIP) (Klein et al., 2012). We present new paleomagnetic and U-Pb geochronological results for the Santa Rosa and Sobreiro Formations in Pará state, southern Amazonian craton. Our new results for ~1880 Ma help to better define the APW path between 1960 Ma and 1860 Ma for the Amazonian craton as well as to test the possible occurrence of TPW during 1880–1860 Ma.

2. Geological setting and lithology

The Amazonian craton (~4,400,000 km^2) is one of the largest cratons in the world (Almeida et al., 1981). It includes the Guiana Shield in the north and the Central-Brazil (or Guaporé) Shield in the south, bisected by the Amazon sedimentary basin (Santos et al., 2000; Schobbenhaus et al., 1984) (Fig. 1). Current models for its Precambrian geodynamics primarily based on geochronological data recognize six (Tassinari and Macambira, 1999; Tassinari and Macambira, 2004) or seven/eight (Santos and Tassinari, 2003; Vasquez et al., 2008) geotectonic provinces. In this work, we follow the model of Tassinari and Macambira (1999) and Tassinari and Macambira (2004), that is used by several other authors (Cordani and Teixeira, 2007; Cordani et al., 2009; Schobbenhaus et al., 2004). Two ~2500 Ma Archean nuclei are formed by gneissic-granitoid terrains and greenstone belts (Central Amazonian Province) exposed in the southeast of Roraima, northeast of Amazonas, and northwest of Pará states (Cordani and Teixeira, 2007). These nuclei were reworked by Paleoproterozoic events (2260–2050 Ma) forming the Maroni-Itacaiúnas Province to the northeast (Tassinari and Macambira, 1999; Tassinari et al., 2000).

One of the most striking characteristic in the Amazonian craton is the abundant A-type intraplate magmatism between 2000 and 1860 Ma (Brito Neves, 2011). The older ~1960 Ma large plutonic and volcanic province includes the "Surumu" magmatism in the Central Amazonian Province in Roraima. The volcanic and plutonic rocks with ages between 1890 and 1860 Ma have generally been referred to as the "Uatumã event" (*sensu stricto*). The associated plutonism (Serra dos Carajás, Cigano, Seringa, Jamon, Musa, Redenção granites, *etc.*) has been studied in detail (See Dall'Agnol et al., 2005 for a review) in the Archean portion of the Central-Brazil Shield. The corresponding volcanic units include the Iriri, Santa Rosa and Iricoumé Formations in northern Brazil. The extension of the Uatumã event may exceed an area larger than 1,500,000 km^2. It forms a 1400 km long NW magmatic belt, considered as a SLIP (Klein et al., 2012). It is worth noting that two younger large intraplate magmatic events also affected the Amazonian craton: the ~1790 Ma "Avanavero-Crepori" event and the intrusion of an AMCG (anorthosite-mangerite-charnockite-granite) suite between 1600 and 1400 Ma (Sadowski and Bettencourt, 1996; Vigneresse, 2005). The deposition of the ~1870 Ma sedimentary rocks from the Roraima Supergroup in northern Brazil and Venezuela emphasizes the stability of the Amazonian craton (Santos et al., 2003). To the southwest the Paleo-Mesoproterozoic evolution is marked by the successive accretion of subduction-related juvenile magmatic arcs: the 1980–1810 Ma Ventuari-Tapajós, the 1780–1600 Ma Rio Negro-Juruena, and the 1550–1300 Ma Rondonian-San Ignacio Provinces (Cordani and Teixeira, 2007; Pinho et al., 2003; Schobbenhaus et al., 2003; Tassinari et al., 2000), which culminated with the collision of the Paraguá Terrane at *ca.* 1320 Ma (Bettencourt et al., 2010; Rizzotto and Hartmann, 2012).

The study area (Fig. 1) is located near to the São Felix do Xingu City in the Rio Maria domain, which corresponds to the southern portion of the Carajás Province (Santos and Tassinari, 2003). This domain is composed by the Archean nucleus of the Central-Brazil Shield in the Central Amazonian Province (Cordani and Teixeira, 2007), SE of the Pará State. The São Felix do Xingu region is about 300 km far from ~560 to 520 Ma Araguaia Belt, a Neoproterozoic to Cambrian Brasiliano orogenic system at the eastern border of the Amazonian craton (Fig. 1). The Archean TTGs (~2700 Ma Rio Maria granodiorite) and metavolcano-sedimentary units of the Itacaiúnas Supergroup are exposed only to the east of Xingu River (Araújo et al., 1988). Well-preserved ~1880–1860 Ma volcanic and plutonic units belonging to the Uatumã SLIP occur across the area. It comprises the 1880 Ma A-type felsic Santa Rosa Formation and the underlying calk-alkaline mafic Sobreiro Formation, which are crosscut by the 1860 Ma A-type tin-bearing granitoid of the Velho Guilherme Suite (Teixeira et al., 2002). The Sobreiro and Santa Rosa Formations were studied in detail by Fernandes et al. (2011) and Juliani and Fernandes (2010). These rocks have been altered by hypogenic

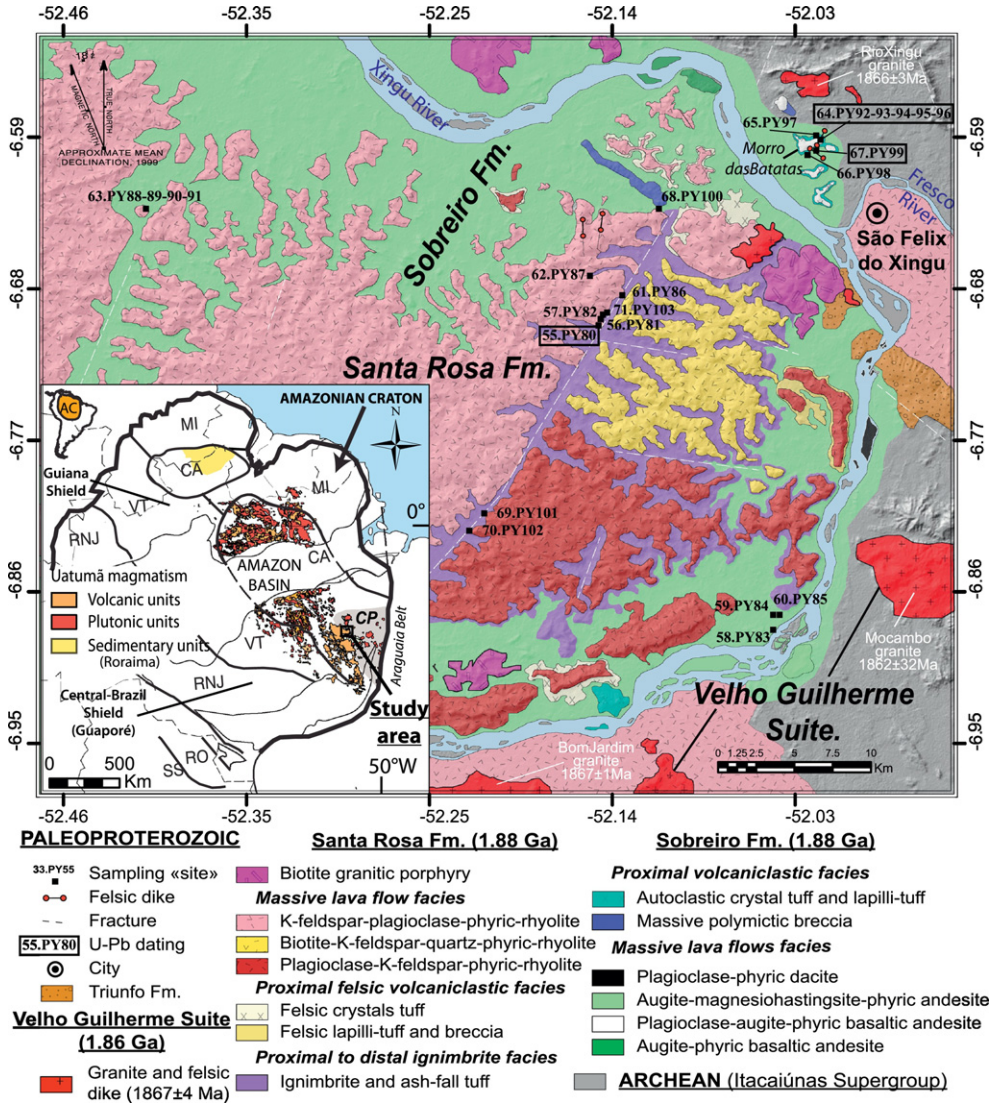


Fig. 1. Geological map of São Felix do Xingu region with location of sampling sites (full square) for Santa Rosa and Sobreiro Formations (adapted from Juliani and Fernandes (2010)). Inset: The ~1880 Ma volcano-plutonic Uatumã event and the main tectonic provinces of the Amazonian craton (adapted from Cordani and Teixeira (2007)). *Abbreviations*: CP = Carajás Province (in grey); CA = Central Amazonian Province; MI = Maroni-Itacaiunas Province; VT = Ventuari-Tapájos Province; RNJ = Rio Negro-Juruena Province; RO = Rondonian-San Ignácio Province; SS = Sunsás province.

hydrothermal fluids (da Cruz et al., 2015; da Cruz et al., 2016). A brief petrographic review for the Sobreiro and Santa Rosa formations is provided below.

The Sobreiro Formation is the basal unit of the Uatumã event with a flat topography along the Xingu River, but its relation with the Santa Rosa Formation is not well defined (Roverato, 2016). The Sobreiro Formation comprises basaltic andesite, andesite and less dacite massive lava flows and volcanoclastic rocks with high-K calk-alkaline signature (Fernandes et al., 2011; Roverato et al., 2017). According to da Cruz et al. (2016) late- to post-magmatic hydrothermal alteration in these rocks is responsible for a secondary paragenesis characterized by epidote, chlorite, carbonate, clinozoisite, sericite, quartz, albite, hematite and pyrite. A prehnite-pumpellyite association is common in the Sobreiro Formation. An age of 1880 ± 6 Ma for this unit was obtained on a dacite by Pb-Pb evaporation on zircon (Pinho et al., 2006).

The Santa Rosa Formation is a thick felsic volcanic formation that usually occupies the topographic highs in the region. Juliani and Fernandes (2010) distinguished four lithological facies: (i) a rhyolitic lava flow and thick dikes of banded rhyolite and ignimbrite, (ii) highly rheomorphic felsic ignimbrite associated with unwelded ash tuff, (iii) felsic crystal tuff, lapilli-tuff, and co-ignimbritic breccias, (iv) granitic porphyry stocks and dikes, and subordinate equigranular granitic intrusions. Two Pb-Pb evaporation ages on zircon of 1884 ± 1.7 Ma and 1879 ± 2 Ma, respectively obtained from an ash tuff and a rhyolite of the Santa Rosa Formation, are reported in Juliani and Fernandes (2010). They interpret the emplacement of these felsic lavas as a dome-shaped structure controlled by two major NE-SW lineaments related to several intrusive episodes. The rocks show an A-type geochemical signature associated with a peraluminous character. They show K-feldspar, euhedral plagioclase, quartz phenocrysts, rare biotite, in a felsic fine-grained groundmass. Spherulitic and granophyric textures are present. Primary accessories are zircon, fluorite, titanite, Fe-Ti oxides, and apatite. The ignimbrites show a welded flow-like eutaxitic banded texture. Granite porphyries are massive with a reddish-pink color with K-feldspar, plagioclase, and quartz phenocrysts. Like the Sobreiro Formation, the Santa Rosa Formation shows hydrothermal alteration with secondary minerals such as microcline, sericite, quartz, carbonate, chlorite, gold, clay minerals and Fe-oxides.

The Velho Guilherme Suite (VGS) is represented in the area by plutonic bodies (Antônio Vicente, Serra da Queimada, Rio Xingu, Mocambo, Bom Jardim granites) and associated dikes. These units cut across the volcanics of Santa Rosa and Sobreiro Formations. The Antônio Vicente,

Mocambo and Rio Xingu granites are dated at 1867 ± 4 Ma, 1862 ± 16 Ma and 1866 ± 3 Ma, respectively, by Pb-Pb evaporation on zircon (Teixeira et al., 2002).

The geodynamic context for the emplacement of these volcanic rocks is not well established yet. It has been attributed to a large scale extensional magmatic event in the Amazonian craton, hence the notion of SLIP (Ferreira and Lamarão, 2013; Klein et al., 2012). An alternative model involves the migration of a continental magmatic arc from the Tapajós Province to the Xingu Region due to the presence of a E-W flat subduction located to the Serra do Cachimbo region, in the southwestern of the continental plate (Juliani et al., 2009). The A-type nature of most volcanic and granitic rocks favors the intraplate scenario (da Silva et al., 2016).

3. Sampling and analytical methods

3.1. Paleomagnetism

The São Felix do Xingu region was selected for paleomagnetic sampling because it is one of the most studied areas in the Carajás Province with precise petrology and owns an exceptional amount of *in situ* outcrops (da Cruz et al., 2016; Fernandes et al., 2011; Juliani and Fernandes, 2010). We sampled 18 sites (Table 1; Fig. 1): (i) seven rhyolite lava flows sites (Sites 55, 57, 61, 62, 63, 69, 70), 2 ignimbrite sites (Sites 56, 71), 1 felsic dike (Site 67) and 1 volcanic breccia site (Site 68) from the Santa Rosa Formation, (ii) 6 andesite sites (Sites 58, 59, 60, $64_{\text{samples PY96D-R}}$, 65, 66) from the Sobreiro Formation, (iii) 1 felsic dike from the Velho Guilherme Suite (Site $64_{\text{PY92-96A-C}}$). A total of 142 cylindrical samples (~2.54 cm in diameter) and 7 blocks were collected and oriented using in most cases a sun compass and a magnetic compass. Due to the challenging conditions of outcrops (jungle, mature soils), the lithological contacts are difficult to observe in the region. Nevertheless, it was possible to perform a baked contact test (Buchan et al., 2007) at site 64 where a composite NE-trending felsic dike of the Velho Guilherme Suite intrudes a volcanoclastic deposit (andesitic composition) of the Sobreiro Formation at Morro das Batatas (Fig. 1). The dike is a ~10 m thick composite dike with a complex ramification. A 30 cm thick chilled margin shows that the dike was emplaced into cooler rocks. Three cylindrical cores ($PY96_{A-B-C}$) and 4 oriented blocks (PY92-93-94-95 in Table 1) were sampled from the dike. The host andesite from the Sobreiro Formation is visible across >100 m after the contact which allowed sampling 14 cylindrical cores at several distances from the contact: three at 0–5 cm (PY96D, E, R), three at 20 cm (PY96F, G, H), one at 1 m (PY96J), one at 3 m (PY96K), one at 6 m (PY96L), one at 10 m (PY96M), and four at ~20 m (PY96N-Q).

Preparation and analysis of the samples were performed in the Laboratório de Paleomagnetismo - Universidade de São Paulo (USPmag). Extracted cylindrical cores in the field and from sampled blocks were cut in 2.2 cm high standards specimens (total of 505). Magnetic susceptibility and anisotropy of magnetic susceptibility (AMS) were measured using a KLY-4S Kappabridge (AGICO, Czech. Republic) with a sensitivity of 2×10^{-8} SI. In order to isolate a characteristic remanent magnetization (ChRM), conventional stepwise alternating field (AF) and thermal demagnetization were performed in a magnetically-shielded room with ambient field <1000 nT. Steps of 2.5 mT (up to 15 mT) and 5 mT (15–100 mT) were selected for AF demagnetization. Devices used were a tumbler Molspin AF demagnetizer coupled to a JR-6A spinner magnetometer (AGICO, Czech Republic) to measure the remanence, an automated three-axis AF demagnetizer coupled to a horizontal 2G-Enterprises™ DC-SQUID magnetometer, and an AF demagnetization coils coupled to a vertical 2G-Enterprises™ DC-SQUID magnetometer with an automatic RAPID sample changer (Kirschvink et al., 2008). Thermal demagnetization comprised steps of 50 °C (until 500 °C) and 20 °C (until 700 °C) using a TD-48 furnace (ASC Scientific). A combination of AF and thermal cleaning was carried out for specimens with a high coercivity component. Principal component analysis (Kirschvink, 1980) was used to determine the remanence directions using orthogonal vector diagrams (Zijderveld, 1967). Only vectors

Table 1
Paleomagnetic sampling in São Felix do Xingu region.

Site	Sample	Localization	Lithology	Geochronology
Santa Rosa Formation				
55	PY80	6.7°S/307.85°E	Rhyolite lava flow	1877.4 ± 4.3 Ma U-Pb zrn (this study)
56	PY81	6.69°S/307.85°E	Ignimbrite	
57	PY82	6.69°S/307.85°E	Rhyolite lava flow	
61	PY86	6.68°S/307.87°E	Rhyolite lava flow	
62	PY87	6.67°S/307.85°E	Rhyolite lava flow	
63	PY88-89-90-91	6.63°S/307.59°E	Rhyolite lava flow	
67	PY99	6.6°S/307.98°E	Felsic microgranite dike - coarse grained	1895 ± 11 Ma U-Pb zrn (this study)
68	PY100	6.63°S/307.89°E	Volcanoclastic breccia	
69	PY101	6.81°S/307.79°E	Rhyolite lava flow	
70	PY102	6.82°S/307.78°E	Rhyolite lava flow	
71	PY103	6.69°S/307.86°E	Ignimbrite	
Sobreiro Formation				
58	PY83	6.88°S/307.96°E	Volcaniclastic deposit (andesitic)	
59	PY84	6.87°S/307.96°E	Volcaniclastic deposit (andesitic)	
60	PY85	6.87°S/307.96°E	Volcaniclastic deposit (andesitic)	
	$PY96_{D-R}$	6.59°S/307.98°E	Volcaniclastic deposit (andesitic)	
64[a]	$PY96_{D, E, R}$	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: <5 cm at contact	1880 ± 6 Ma Pb-Pb zrn (Pinho et al., 2006)
	$PY96_{F, G, H}$	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: 5–20 cm from the contact	
	PY96J	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: 1 m from the contact	
	PY96K	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: 3 m from the contact	
	PY96L	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: 6 m from the contact	
	PY96M	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: 10 m from the contact	
	$PY96_{N-Q}$	6.59°S/307.98°E	Volcaniclastic deposit (andesitic) - *BCT*: 20 m from the contact	
65	PY97	6.59°S/307.98°E	Volcaniclastic deposit (andesitic)	
66	PY98	6.6°S/307.98°E	Volcaniclastic deposit (andesitic)	
Velho Guilherme Suite				
64[a]	$PY92–PY93–PY96_{A-C}$	6.59°S/307.98°E	Felsic microgranite dike - fine grained (chilled margin)	1853.7 ± 6.2 Ma U-Pb zrn (this study)
	PY94–PY95	6.59°S/307.98°E	Felsic microgranite dike - coarse grained	

[a] Site sampled for the baked contact test (BCT) - distances from the contact are indicated.

with mean angular deviation (MAD) smaller than 8° were considered. For some sites, remagnetization great circles analysis (Halls, 1978) was also employed to determine high coercivity/high-blocking temperature components. Site-mean directions and paleomagnetic poles were calculated using Fisher's statistics (Fisher, 1953). Paleomagnetic data processing was carried out using REMASOFT, GMAP, and Super-IAPD software (Chadima and Hrouda, 2006; Torsvik et al., 2000; Torsvik and Smethurst, 1999). Velocity of the different cratons between 2100 and 1200 Ma was calculated using PMTec (Wu et al., 2015) with 20 Ma time windows at 5 Ma steps. Paleomac (Cogné, 2003) and GPlates were used for great circle calculations and paleogeographic reconstructions (Boyden et al., 2011).

The magnetic mineralogy was investigated on selected specimens to determine the carriers of magnetic remanence. Hysteresis loops and isothermal remanent magnetization (IRM) curves were determined using a MicroMag-VSM Model 3900 (Princeton Measurements Corporation). Thermomagnetic experiments (susceptibility *versus* temperature) were conducted in argon atmosphere in low and high temperature conditions using a CS-4 apparatus coupled to the KLY-4S Kappabridge instrument (AGICO, Czech Republic). Thin and polished sections were studied under transmitted and reflected light microscopy and scanning electron microscopy (SEM) for mineralogy and texture analysis.

3.2. Geochronology

We selected samples from representative paleomagnetic results for U-Pb age determination to upgrade previously published Pb-Pb ages for the same formations: (i) one rhyolite (PY80B2) and one associated dike (PY99) from Santa Rosa Formation, (ii) one dike (PY92B1) from the Velho Guilherme Suite. Before analysis, zircons were characterized by cathodoluminescence (CL) and back-scattered electron (BSE) images using scanning electron microscopy (SEM) (Fig. 2).

Two different dating techniques were used in this study: (i) zircon crystals from one dike of the Santa Rosa Formation (PY99) were analyzed by the Sensitive High Resolution Ion Microprobe (SHRIMP) at the Centro de Pesquisas Geocronológicas (CPGeo) of the Instituto de Geociências, Universidade de São Paulo (IGc-USP), (ii) zircon crystals from the Santa Rosa lava flow (PY80B2) and of a Velho Guilherme dike (PY92B1) were analyzed *in-situ* on polished sections by Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS) at the Universidade Federal de Ouro Preto (UFOP), Minas Gerais State, Brazil.

SHRIMP analyzes were performed during one session using a SHRIMP IIe/MC. The primary beam was set with Kohler aperture = 120 μm, spot size = 30 μm, and O-2 beam density = 2.5–7 ηA (dependent of brightness aperture), and the secondary beam was set with source slit = 80 μm; mass resolutions for 196 (Zr_2O), ^{206}Pb, ^{207}Pb, ^{208}Pb, ^{238}U, 248 (ThO) and 254 (UO) ranging between 5000 and 5500 (1%), and residues < 0.025. Data were reduced with SQUID 1.06 software, and Concordia diagrams were plotted with ISOPLOT 4 (Ludwig, 2009). Common lead corrections usually use ^{204}Pb according to Stacey and Kramers (1975). Temora 2 standard was used as $^{206}Pb/^{238}U$ ages reference every ten analyses (416.78 Ma, Black et al. (2004)) and provided a weighted mean of standard deviation in Pb/U of 0.003524 (1σ, n = 19, MSWD = 8.60).

ICP-MS analyses were performed during two sessions using a ThermoScientific Element 2 coupled to a LSX-213 G2 laser (CETAC Technologies) with a 20 μm laser spot size. The data were reduced with the software Glitter (Van Achterbergh et al., 2001) and ages were calculated using the ISOPLOT 4 (Ludwig, 2009) program with dating uncertainties at the 2 sigma level. The GJ-1 standard zircon (608.5 ± 0.4 Ma by ID-TIMS) (Jackson et al., 2004) was used for the calibration of the U-Pb analysis in association with the Plešovice secondary standard (337.13 ± 0.37 Ma by ID-TIMS) (Sláma et al., 2008) to test the accuracy of the results each ten analysis. GJ-1 standard zircon gave Concordia ages of 607.00 ± 0.91 Ma (2σ, n = 29, MSWD = 0.42) and 606.1 ± 1.6 Ma (2σ, n = 11, MSWD = 4.3) for the two sessions, respectively. The Plešovice secondary standard gave Concordia ages of 336.80 ± 0.73 Ma (2σ, n = 13, MSWD = 0.87) and 337.6 ± 3.3 Ma (2σ, n = 7, MSWD = 3.4) for the two sessions, respectively. Calculated ages for the two standards are consistent with the reference values.

4. U-Pb geochronology

4.1. Zircon morphology

Zircons from the Santa Rosa rhyolite sample PY80B2 are associated with oxides (martite – Fe_2O_3, Mt). They are euhedral to subeuhedral with typical growth zoning in cathodoluminescence (CL) images pointing to a magmatic crystallization (Fig. 2-A). Zircons separated from the dike of the same unit (sample PY99) are euhedral 250 μm in size and also show growth zoning and inclusions (Fig. 2-C). Zircons from the Velho Guilherme Suite dike (PY92B1) show a range of morphologies from euhedral to subeuhedral with different types of magmatic zoning and usually contain apatite inclusions suggesting hydrothermal processes during the formation of these rocks (Fig. 2-B). Zircons studied here, show features of magmatic crystallization without overgrowth and recrystallization, suggesting a magmatic origin.

4.2. U-Pb geochronology results

Eight zircons from the rhyolite flow (PY80B2) show ages between 1890 and 1867 Ma (Table 2). An upper intercept of 1877.4 ± 4.3 Ma (MSWD = 1.7, probability = 0.099) was calculated for this sample that likely represents the emplacement age of the Santa Rosa rhyolite (Fig. 2-A). It was also possible to calculate a Concordia age of 1879.6 ± 7.4 Ma (MSWD = 5.2, probability = 0.023) using a cluster of four zircons. The weighted average age for the eight zircons is 1877 ± 14 Ma (MSWD = 0.22, probability = 0.98). The other sample from the Santa Rosa Formation corresponds to the PY99 dike for which SHRIMP analyses were performed. For this sample, nine analyzed zircons yielded an upper intercept of 1895 ± 11 Ma (MSWD = 1.7), which agrees with the weighted age of 1896 ± 17 Ma (MSWD = 1.7) (Fig. 2-C). These results are *ca.* 20–40 Ma older than the NE-trending dike from the Velho Guilherme Suite (see below), confirming it must belong to the vein system that originated in the Santa Rosa Formation.

The ages for the NE-trending dike of the Velho Guilherme Suite (PY92B1) range between 1867 and 1841 Ma. These ages are slightly younger than those from the rhyolite PY80B2 and from the PY99 felsic dike (Table 2). The upper intercept calculated using eight zircons is 1853.7 ± 6.2 Ma (MSWD = 1.2, probability = 0.30) (Fig. 2-B). Recent Pb loss or alteration of zircon grains resulted in three discordant ages. The remaining five concordant zircon ages were used to calculate a Concordia age of 1853.3 ± 3.8 Ma (MSWD = 1.8, probability = 0.18). The weighted average age is 1852 ± 15 Ma (MSWD = 0.20, probability = 0.99).

5. Paleomagnetic results

Rocks from the São Felix do Xingu area show three characteristic components (ChRM) after thermal and AF treatment, with thermal treatment being generally more efficient than AF demagnetization (Fig. 3). Thermal demagnetization provided better resolution of the trajectories due to the hematite.

Component SF1 was retrieved from eight sites of the Santa Rosa Formation and two sites of the Sobreiro Formation. It corresponds to a stable, steep positive (negative) inclination ChRM isolated after AF and thermal demagnetization (Fig. 3-A–D, F). This ChRM direction has unblocking temperatures between 620 °C and 700 °C, the magnetic vector always moving toward the origin in the orthogonal diagram. NRM intensities vary between 0.59 A/m and 1.44 A/m, which are common values for rhyolitic rocks carrying high coercivity/blocking temperature minerals such as hematite (Dunlop and Ozdemir, 1997). Four sites show

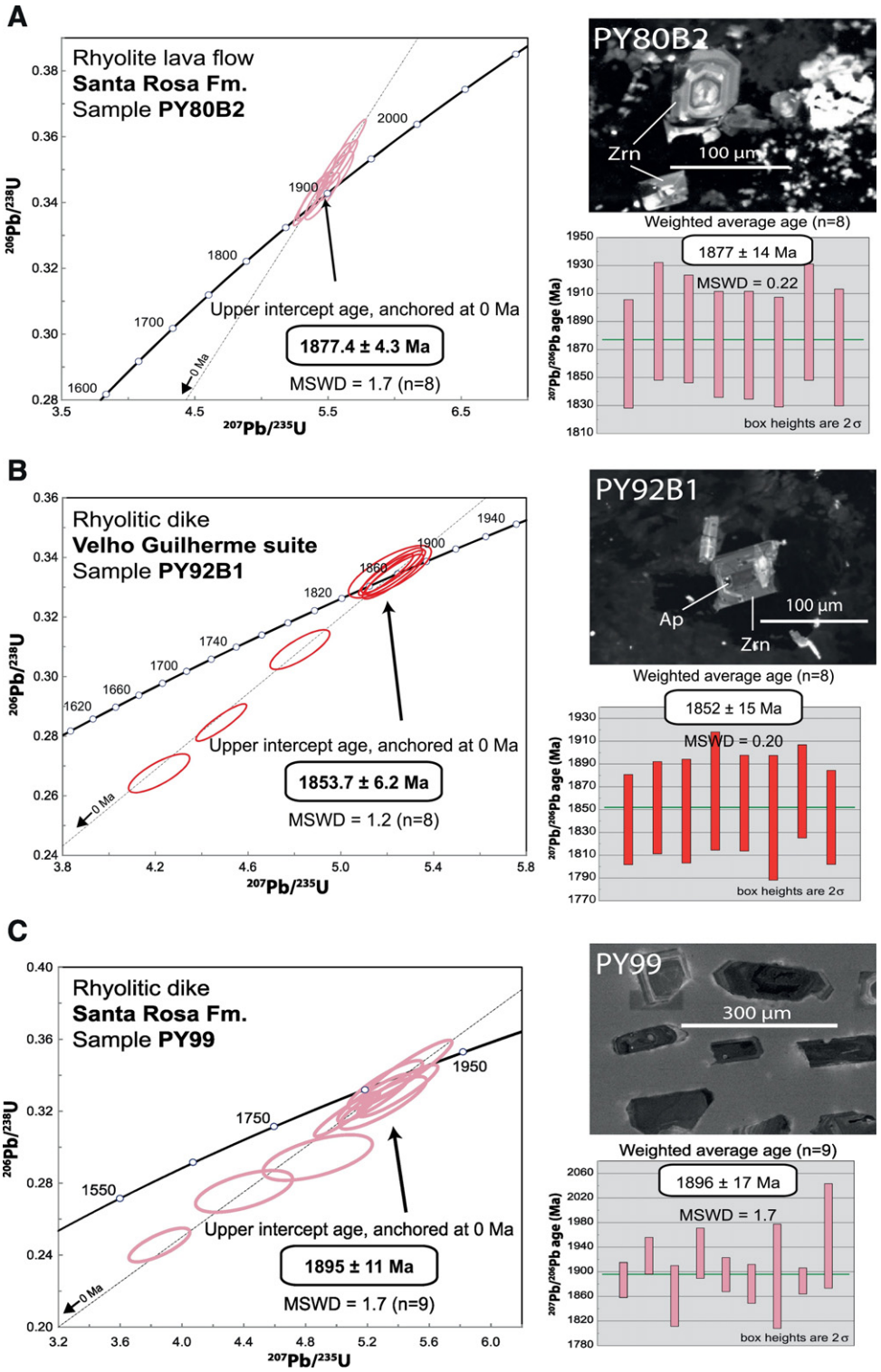


Fig. 2. (A) Concordia diagrams showing U-Pb zircon LA-ICP-MS results for the rhyolite lava flow sample (PY80B2) and cathodoluminescence (CL) images of typical zircon grains for the sample with the weighted average age. (B) Concordia diagrams showing U-Pb zircon LA-ICP-MS results for the sample of the NE-dike (PY92B1) and cathodoluminescence images of typical zircon grains for the Velho Guilherme Suite with the weighted average age. (C) Concordia diagrams showing U-Pb zircon SHRIMP results for the NW-dike (PY99) and cathodoluminescence images of typical zircon grains for the dike of the Santa Rosa Formation with the weighted average age.

'normal' polarity and six sites a 'reverse' polarity so that a reversal test was performed (McFadden and McElhinny, 1990). Nevertheless, despite the similar mean direction for 'normal' (N) and 'reverse' (R) polarities, the test resulted indeterminate, probably due to the small number of sites (10). To circumvent this problem, we used specimen's directions to calculate mean directions for the 'normal' and 'reverse' polarities: Dm = 311.6°, Im = −80.0° (N = 59, R = 55.82) and Dm = 92.7°, Im = 84.6° (N = 29, R = 27.93), respectively. The reversal test now resulted in a critical gamma of 7.3°, which is greater than the angle between the normal and reversed axes. So, it can be considered as positive and classified with quality "B" according to McFadden and McElhinny (1990). Using the specimen directions, the mean SF1 component is Dm =

Table 2
Zircon U-Pb data obtained by *in situ* laser ablation ICP-MS and SHRIMP.

Analysis	Th/U	Isotopic ratios						ρ	Ages (Ma)				Conc. (%)
		$^{207}Pb/^{206}Pb$	±1σ	$^{207}Pb/^{235}U$	±1σ	$^{206}Pb/^{238}U$	±1σ		$^{207}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	
Rhyolite lava flow - Santa Rosa formation (PY80B2) - LA-ICP-MS													
PY80B2 #021	0.43	0.1142	0.0012	5.3573	0.0478	0.3399	0.0028	0.94	1867	19	1886	14	100.4
PY80B2 #036	0.46	0.1157	0.0014	5.5540	0.0559	0.3480	0.0031	0.87	1890	21	1925	15	100.8
PY80B2 #039	0.45	0.1153	0.0012	5.5967	0.0511	0.3517	0.0030	0.94	1885	19	1943	14	101.4
PY80B2 #040	0.31	0.1146	0.0012	5.6628	0.0498	0.3579	0.0030	0.96	1874	19	1972	14	102.4
PY80B2 #043	0.42	0.1146	0.0012	5.5412	0.0520	0.3505	0.0031	0.94	1873	19	1937	15	101.5
PY80B2 #044	0.37	0.1143	0.0013	5.3855	0.0487	0.3414	0.0029	0.92	1868	20	1894	14	100.6
PY80B2 #045	0.45	0.1156	0.0013	5.4570	0.0529	0.3419	0.0029	0.88	1890	21	1896	14	100.1
PY80B2 #047	0.47	0.1145	0.0013	5.4090	0.0528	0.3423	0.0030	0.88	1872	21	1898	14	100.6
NE-Dike - Velho Guilherme suite (PY92B1) - LA-ICP-MS													
PY92B1 #027	1.35	0.1126	0.0012	5.1945	0.0482	0.3342	0.0028	0.91	1841	20	1859	14	100.4
PY92B1 #028	1.00	0.1132	0.0013	5.2353	0.0530	0.3349	0.0031	0.91	1852	20	1862	15	100.2
PY92B1 #029	1.23	0.1130	0.0014	4.8234	0.0526	0.3091	0.0027	0.80	1849	23	1736	13	96.9
PY92B1 #031	0.54	0.1142	0.0017	4.2162	0.0539	0.2676	0.0027	0.78	1867	26	1528	13	90.3
PY92B1 #036	0.78	0.1135	0.0013	5.2250	0.0540	0.3336	0.0031	0.89	1856	21	1856	15	99.9
PY92B1 #037	0.69	0.1127	0.0017	5.2117	0.0730	0.3352	0.0036	0.76	1843	27	1864	17	100.5
PY92B1 #038	0.64	0.1141	0.0013	4.4842	0.0445	0.2846	0.0026	0.90	1866	20	1614	13	93.0
PY92B1 #051	1.02	0.1127	0.0013	5.2209	0.0538	0.3356	0.0031	0.89	1843	21	1866	15	100.5
NW-Dike - Santa Rosa formation (PY99) - SHRIMP													
PY99 #1.1	1.12	0.1155	0.7900	5.4529	2.2325	0.3425	2.0881	0.94	1887	14	1899	34	100.7
PY99 #3.1	0.97	0.1180	0.8338	5.3993	1.9633	0.3318	1.7775	0.91	1926	15	1847	29	95.3
PY99 #5.1	0.73	0.1138	1.3683	3.8580	2.0808	0.2458	1.5677	0.75	1861	25	1417	20	73.4
PY99 #6.1	1.12	0.1183	1.1394	5.3020	2.1674	0.3251	1.8437	0.85	1931	20	1814	29	93.1
PY99 #7.1	1.17	0.1160	0.7687	5.2441	1.8267	0.3278	1.6571	0.91	1896	14	1828	26	95.9
PY99 #9.1	1.15	0.1151	0.8780	5.3175	1.8610	0.3352	1.6409	0.88	1881	16	1863	27	98.9
PY99 #10.1	1.61	0.1159	2.3562	4.4019	2.9168	0.2756	1.7193	0.59	1893	42	1569	24	80.7
PY99 #12.1	0.66	0.1154	0.5879	5.0709	1.6963	0.3188	1.5912	0.94	1886	11	1784	25	93.8
PY99 #13.1	1.18	0.1202	2.3784	4.8810	2.9597	0.2946	1.7615	0.60	1959	42	1664	26	83.0

123.6°, Im = 81.8° (N = 88, α_{95} = 3.5°, k = 19.9). Site mean directions for the 10 sites (N and R polarity) cluster around the mean Dm = 149.2°, Im = 79° (a95 = 9.6°, k = 26.4), yielding to a SF1 paleomagnetic pole of Plat = 24.7°S and Plong = 319.7°E (A_{95} = 16.9°) (Table 3 – Fig. 4).

Component SF2 is a southwest/northeast and low inclination direction present in 15 analyzed sites. For rhyolites of the Santa Rosa Formation this component is may be found together with component SF1 (Fig. 3-A, B, F). It is also observed in four sites of the Sobreiro andesites (Fig. 3-D), and in the felsic dike of the Velho Guilherme Suite, where it was isolated by the AF treatment (Fig. 3-E). This component is always characterized by unblocking temperatures between 520 and 580 °C, and coercivities between 10 and 100 mT, suggesting it is carried by magnetite, in contrast with component SF1, which is normally carried by hematite. A baked contact test for this component is presented hereafter for site 64. The mean site for the SF2 component is Dm = 239.2°, Im = 10.6° (N = 15, α_{95} = 12.4°, k = 10.5), yielding to a paleomagnetic pole of Plat = 31.1°S and Plong = 220.1°E (A_{95} = 9.7°) (Fig. 4).

Component SF3 is a northerly direction with positive inclination (Fig. 3-C). It was isolated in the NW-trending dike of the Santa Rosa Formation, well-dated at 1895 ± 11 Ma. The component SF3 is different from components SF1 and SF2, despite a possible overlapping between SF1 and SF3. It is found in all the studied units, being usually isolated at lower unblocking temperatures (100–200 °C) (Fig. 3-C). The presence of SF3 in 14 sites from different lithologies of different ages suggests that it may be an overprint. The mean for the SF3 direction is Dm = 0.9° Im = 61.5° (α_{95} = 11.9°, k = 12.2), yielding to a paleomagnetic pole of Plat = 40.7°S and Plong = 128.8°E (A_{95} = 15.6°) (Fig. 4).

6. Baked contact test

The baked contact test was performed at site 64 (PY92-93-94-95-96) where a ~10 m thick NE-trending dike of the Velho Guilherme Suite cuts the volcaniclastic Sobreiro andesites (Table 1, Fig. 5). It is a composite dike system well-dated at 1853.7 ± 6.2 Ma (this study). The host andesites yielded a Pb-Pb age of 1880 ± 6 Ma (Pinho et al., 2006) on zircon. Samples from the dike show northeastern directions with low downward inclinations (PY92A1), corresponding to SF2 component (Fig. 5). Baked contact rocks present the same SF2 component as shown, for example, in sample PY96D1-1 collected at 0.2 cm from the contact (Fig. 5). A hybrid zone is represented in sample PY96J1 collected 1 m away from the contact, where a SF2 direction was disclosed between 0 and 15 mT and another more stable component is disclosed at higher coercivities > 50 mT or with unblocking temperatures between 580 and 700 °C. This component, carried by hematite, is similar to the SF1 component isolated in the lava flows of the Santa Rosa Formation. At greater distances from the contact, the component SF1 (580–700 °C) is always observed, as shown by samples PY96-K2 (at 3 m) and PY96P1 (at 20 m in Fig. 5). Hydrothermal alteration randomly affected the host volcanoclastic rock (andesitic). For specimen PY96J1 we observe an intense alteration with chlorite, hematite and even covellite being formed at the expenses of amphibole. In contrast, in specimen PY96K hematite is absent and the main oxides are large magnetite grains showing exsolution of ilmenite.

7. Magnetic mineralogy

IRM curves were acquired for 25 specimens and four representative examples are illustrated in Fig. 6. After unmixing of IRM curves by using cumulative log Gaussian functions (Kruiver et al., 2001), two or three magnetic components of different coercivities were identified. In felsic samples of the Santa Rosa Formation (both rhyolites and ignimbrites) two components are generally identified (Fig. 6-A). In all samples of rhyolites and ignimbrites the main component has coercivities between 400 and 1122 mT. This component contributes between 53 and 95% to the total remanence and the values of $B_{1/2}$ and DP (dispersion parameter) are typical of a high coercivity mineral like hematite (Fig. 6-E). The second magnetic component has a $B_{1/2}$ of 32 mT for the ignimbrite

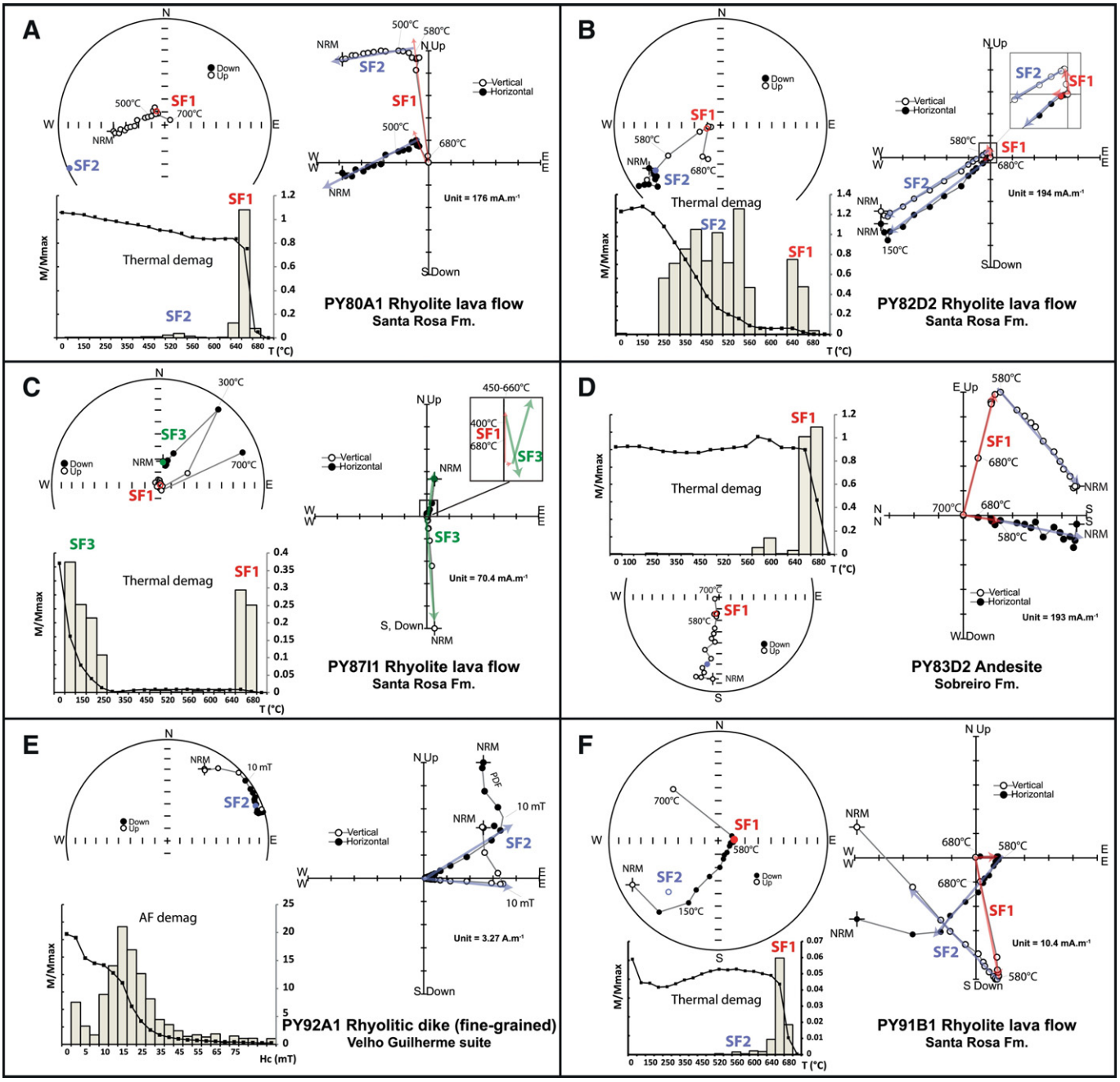


Fig. 3. Examples of AF (alternating field) and thermal demagnetization for samples of the Santa Rosa Formation, Sobreiro Formation and Velho Guilherme Suite. Demagnetization results presented with stereographic projections, orthogonal projections (Zijderveld plot) and normalized magnetization intensity curves (M/M_0 *versus* alternating field H).

PY81E. The contribution of this magnetic component to the total remanence varies between 3 and 43% except for site PY91C where it reaches 87%. The characteristic association between magnetite and hematite is also found in andesites of the Sobreiro Formation (Fig. 6-C). In contrast, sample PY96A from the Velho Guilherme Suite dike (Fig. 6-B) does not contain the high coercivity component, and in the andesite specimen PY96K it is possible to divide the IRM curve into two components "hard magnetite" and "soft magnetite" (Fig. 6-D).

Kruiver's parameters DP (dispersion parameter, width of the distribution of the coercivity) and $B_{1/2}$ can be used to separate the samples according to their magnetic properties (Fig. 6-E). All samples with magnetic behavior dominated by hematite when plotted in Day's diagram (Fig. 7-A) show values of SD (single-domain) hematite according to the Hcr/Hc limit (of 1.17) between SD and MD (multi-domain) hematite established by Özdemir and Dunlop (2014). The hysteresis curve for rhyolite PY82D is wasp-waisted, typical of a mixture of two magnetic phases (Fig. 7-B). It shows a large coercivity indicating the predominance of hematite, which is confirmed by Curie/Néel temperature at around 670–680 °C and a Morin transition ($T_M = -20$ °C) (Fig. 7-C). Most samples with predominance of titanomagnetite fall into the pseudo-single domain (PSD) in the Day's diagram (Fig. 7-D) except for sample PY96Q that plot into the multi-domain (MD) field. These samples are characterized by narrow-waisted hysteresis curves (Fig. 7-E). In thermomagnetic curves the same samples show an abrupt decrease at around 580–600 °C and a sharp increase in susceptibility at the Verwey transition ($Tv = -122$ °C), both indicative of Ti-poor titanomagnetite (Fig. 7-F). A small decrease in the heating curve at *ca.* 410 °C is sometimes observed and could be related to maghemite.

Table 3
n/N – number of analyzed samples/number of samples used in the mean; Dec. – declination; Inc. – inclination; α95, k – Fisher's confidence cone and precision parameter (Fisher, 1953); *G.C* – great circles analysis. VPG – virtual geomagnetic pole; Plong – paleolongitude; Plat – paleolatitude. VGPs values are indicated after switching to the antipodal position.

Site	Sample	Characteristic remanent magnetization (ChRM)					VGP	
		n/N	Dec (°)	Inc (°)	k	α95 (°)	Plat (°N)	Plong (°E)
Component SF1 (~1880 Ma)								
SF1 normal polarity (N)								
55	PY80	13/14	357	−69	154.1	3.3	44.2	130.4
57	PY82	14/14	250.1	70.3	17.2	9.9	−5.8	161.2
61	PY86	10/11	3.2	82.6	97.6	4.9	21.2	127
62	PY87	15/15	328.9	74.6	51.3	5.4	30.8	144.7
Mean sites SF1 N-polarity		4	322.1	78.8	31.7	16.6	23.4	142.2
Mean specimens SF1 N-polarity		59	311.6	−80	18.3	4.4	19.2	143.2
SF1 reversed polarity (R)								
63	PY91	8/15	113.2	86.1	45.2	8.3	9.6	134.8
56	PY81	10/14	313.6	87.2	71.8	5.7	2.8	123.8
71	PY103	5/13	117.1	77.3	*G.C*	2.5	17	150.3
68	PY100	12/17	178.6	58.2	*G.C*	9.5	57.7	129.9
58	PY83	10/17	200.7	65.3	*G.C*	4.8	45.8	107.9
64	PY96	12/33	85.5	67	*G.C*	5.9	2.1	168.2
Mean sites SF1 R-polarity		6	154	79.1	20.4	15.2	25.4	137.9
Mean specimens SF1 R-polarity		29	92.7	84.6	26.2	5.3	7.1	138.7
Component SF2 (~1860 Ma)								
SF2 normal polarity (N)								
64	PY9293949596	8/15	53.7	16.7	19.9	12.7	36.8	43.7
64	PY96	20/33	56.6	3	10.3	10.7	33	31.9
59	PY84	9/11	72	31.6	15.6	13.5	19.2	53.7
65	PY97	6/15	68.4	25.8	16.7	16.9	19.2	21.1
Mean sites SF2 N-polarity		4	62.2	−5	9.7	31.1	27.9	37.2
SF2 reversed polarity (R)								
55	PY80	4/14	254	4	126.1	8.2	16.1	38
57	PY82	12/14	223.8	9.7	8.3	16	46.4	37.9
61	PY86	6/11	262.7	5.5	8.5	24.3	7.5	39.8
62	PY87	2/15	271.3	−4.7	145.4	20.9	−1.5	35.7
63	PY91	7/15	223.1	17.3	10.5	19.6	44.3	18.3
69	PY101	5/11	230.5	34.2	7.3	30.5	39.6	56.5
70	PY102	3/12	227.2	30.1	73.7	14.5	42.9	53.6
56	PY81	3/14	256.8	3.9	13.7	34.7	13.4	38.3
58	PY83	15/17	233.2	25.3	9.8	12.9	37.3	49.5
60	PY85	7/10	200.5	22.9	23.9	12.6	69.2	53.5
66	PY98	3/17	229.2	16.9	357.8	6.5	41.2	43.7
Mean sites SF2 R-polarity		11	238.1	12.7	10.1	15.1	32.3	41.3
Component SF3 (~550 Ma)								
55	PY80	6/14	48.1	57.9	32.9	11.8	26.5	348.5
61	PY86	4/11	313.8	57.9	40	14.7	27.7	268.3
62	PY87	13/15	343	57.5	19.2	9.7	42.5	289.6
63	PY91	4/15	313.1	39.2	95.8	9.4	35.8	251.1
69	PY101	2/11	310.6	59.1	135.2	21.6	24.8	267.9
70	PY102	3/12	10.9	68.1	240	8	31.2	315.7
56	PY81	9/14	350.1	51	61.3	6.6	50.5	294.6
71	PY103	8/13	15	39.6	71.7	6.6	57.4	334.3
64	PY9293949596	6/15	350.6	42.9	20.4	15.2	57.2	292.2
67	PY99	4/4	56.1	55.1	24.2	19.1	22.6	354.9
59	PY84	2/11	15.8	66.1	52.9	35	33	320.4
64	PY96	4/33	11.1	62.5	64.1	11.6	38.6	318.3
65	PY97	5/15	358.8	37	15.2	20.3	62.7	305.5
66	PY98	2/17	57.8	59.7	625.7	10	19.1	350.9

Poles	Mean direction					Paleomagnetic pole	
	N	D_m (°)	I_m (°)	k/K	α95 (°)/A95 (°)	Lat (°N)	Long (°E)
Pole SF1 (N ± R)	10	149.2	79	26.4/9.17	9.6/16.9	−24.7	319.7
Pole SF2 (N ± R)	15	239.2	10.6	10.5/16.46	12.4/9.7	−31.1	220.1
Pole SF3	14	0.9	61.5	12.2/6.6	11.9/15.6	−40.7	128.8

8. Oxide textural analysis

Petrography of rocks in São Felix do Xingu region is well-known (da Cruz et al., 2016) so we limited our analyses to samples representing different paleomagnetic behavior focusing on iron oxides. The most common iron oxides in rhyolites and ignimbrites for the Santa Rosa Formation are primary titanomagnetite with typical sizes of 10–250 μm containing ilmenite lamellae (Fig. 8-A and B). Fig. 8-C shows fine (~ 2 μm) subsolidus exsolution of ilmenite as trellis-lamellae parallel to (111) planes corresponding to the C2–C3 oxidation stages of Haggerty (1991). Sandwich lamellae and some homogeneous oxides have also been observed.

In rhyolites, titanomagnetite is commonly altered to hematite (Martite, Mt) characterized by high anisotropy (Fig. 8-B). This alteration is regarded as the results of late- to post-magmatic oxidation by hydrothermal fluids (Meller et al., 2014; Nédélec and Borisova, 2015; Nédélec et al., 2015). Indeed, the replacement of magnetite by hematite in A-type granitic magmas emplaced at high levels is a common observation

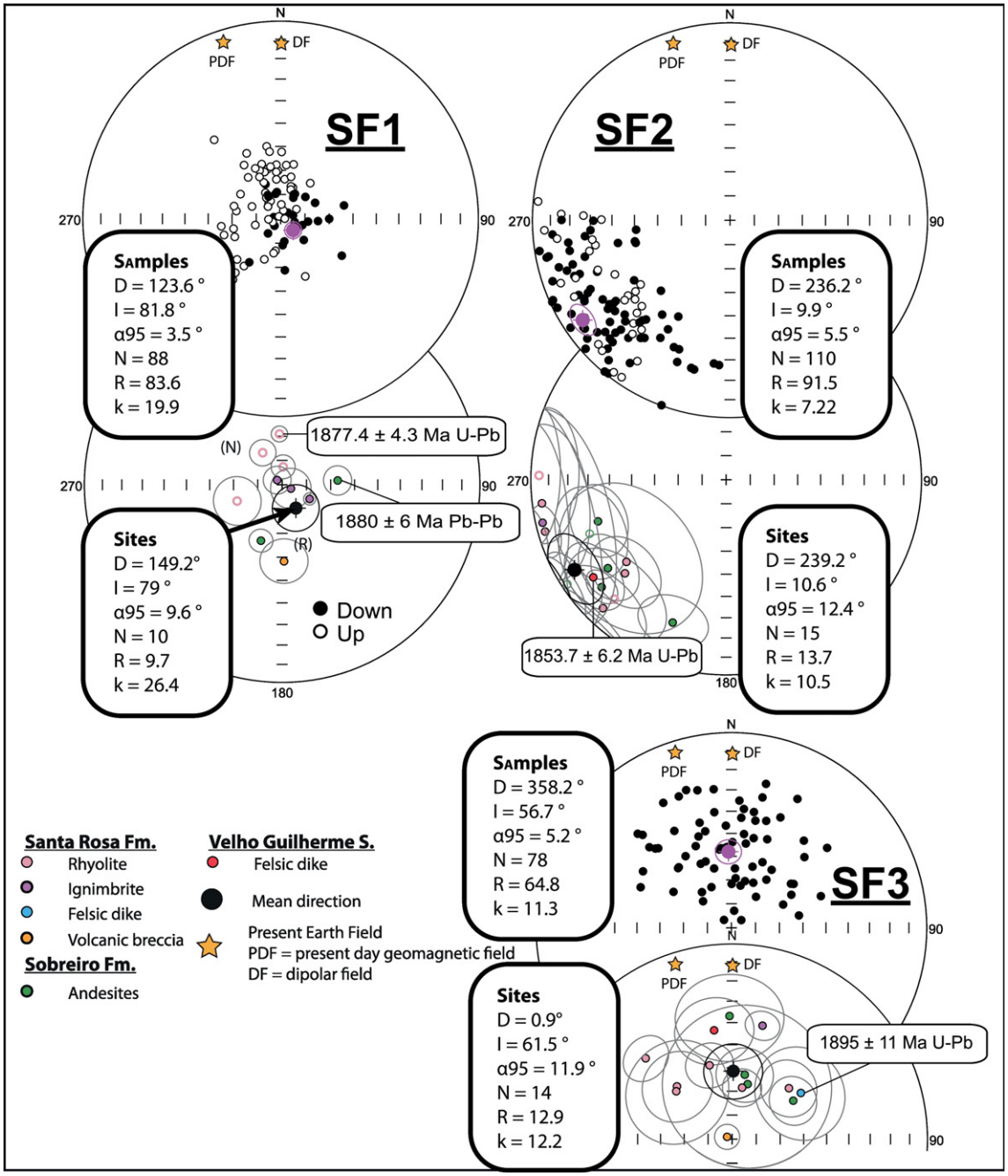


Fig. 4. Stereographic projections of sample and site mean directions for the sites analyzed. Boxes show mean directions and Fisherian statistical parameters for the four mean directions (SF1, SF2 and SF3).

and is regarded as having occurred immediately after emplacement (Nédélec et al., 2015). The recognition of REE-fluoro-carbonate (synchysite) together with fluorine as accessory minerals (Fig. 8-A and D) in Santa Rosa volcanic rocks is evidence for the existence of a F-rich late-stage magmatic fluid (Agangi et al., 2010). This F-bearing fluid of magmatic derivation, possibly mixed with a lower temperature meteoric fluid, was likely responsible for the transformation of magnetite to hematite (Nédélec and Bouchez, 2015).

Spherulites of 0.2–3 cm in size are very common in volcanic facies of Santa Rosa Formation (Fig. 8-D). These features display a radiating texture of quartz and albite that are described as high-temperature crystallization domains (HTCDs) common in silica-rich volcanic rocks (Breitkreuz, 2013). Castro et al. (2008) pointed that the anhydrous mineral assemblage in spherulites crystallized rapidly at the expense of rhyolitic lava or pyroclastic deposits just after emplacement, and was responsible for expulsion of volatile in the magma at the front of the growing spherulites. The diffusion patterns around spherulites support this interpretation and sometimes evidence limited back-diffusion of water after spherulitic growth. Hematite inclusions observed in spherulites are therefore regarded as a magmatic or late-magmatic phase formed during or immediately after spherulitic growth, which rules out the formation of hematite by low-temperature weathering.

Hematite can also be found in association with quartz in veins (Fig. 8-E) as in the site 69.PY101A2 in the plagioclase-K-feldspar-phyric-rhyolite facies. In the dike of the Velho Guilherme Suite, titanomagnetite is also variably altered to hematite (Fig. 8-F). Fig. 8-G shows the alteration of a titanomagnetite with titanite filling ilmenite lamellae. Associated chlorite points to alteration in greenschist facies

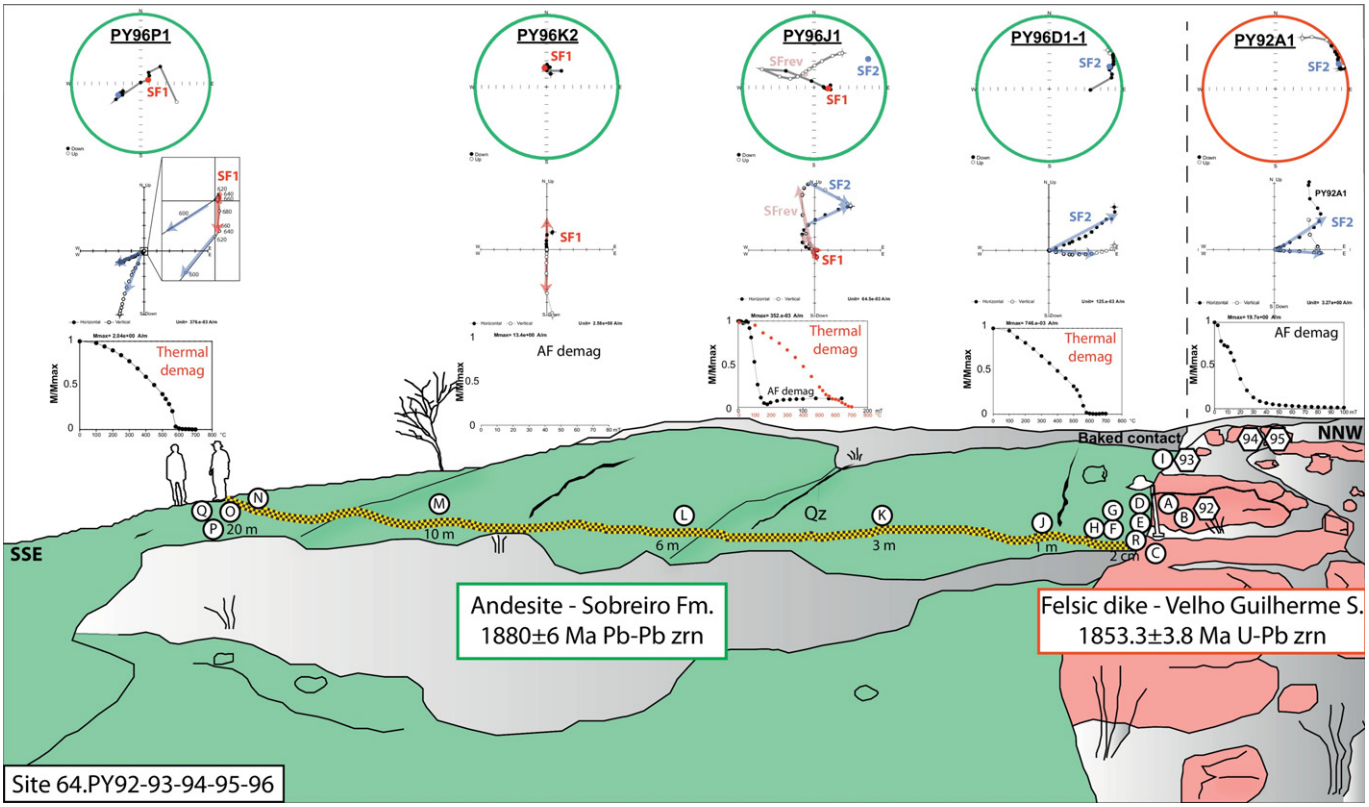


Fig. 5. Baked contact test for a felsic dike of the Velho Guilherme Suite intruding the Sobreiro Formation. The figure shows the locations of the samples and the distance from the contact, stereographic projections, orthogonal projections, and normalized magnetization intensity curves for samples. The different components are illustrated with arrows.

conditions (Fig. 8-G). Close to the contact of the Velho Guilherme dike this alteration is very well marked. The andesite rocks of the Sobreiro Formation (PY85C1) show euhedral amphiboles and plagioclase phenocrysts (Fig. 8-H) with titanomagnetite, rare martite and zircon grains associated.

9. Discussion

9.1. U-Pb geochronology

The U-Pb results presented in this article are consistent with previous Pb-Pb ages obtained on other same units. Juliani and Fernandes (2010) published two Pb-Pb ages on zircons for the Santa Rosa Formation of 1879 ± 2 Ma and 1884 ± 1.7 for a rhyolite and an ash tuff, respectively. We present here the first U-Pb ages on zircons for the Santa Rosa Formation with 1877.4 ± 4.3 Ma for a rhyolite and 1895 ± 11 Ma for a dike. A similar 1880 ± 6 Ma (Pb-Pb) age was also found for andesites from the Sobreiro Formation (Pinho et al., 2006). All geochronological results support a *ca.* 1880 Ma age for the emplacement of these rocks. The age of 1853.7 ± 6.2 Ma obtained here for the sampled Velho Guilherme dike is consistent with previous Pb-Pb ages of 1867 ± 4 Ma, 1862 ± 16 Ma and 1866 ± 3 Ma obtained for the Antônio Vicente, Mocambo and Rio Xingu granites, respectively (Teixeira et al., 2002), and with a SHRIMP age of 1857 ± 8.4 Ma recently reported for another dike from the same Velho Guilherme dike swarm (Roverato, 2016).

9.2. Confidence of the paleomagnetic poles

9.2.1. The ~1880 Ma SF1 component

Eight sites of felsic rocks of Santa Rosa Formation and two sites of the Sobreiro Formation revealed the SF1 remanence component carried by SD hematite (high coercivity and high unblocking temperatures). This hematite is regarded as the result of late- to post-magmatic oxidation by hydrothermal fluids, indicating that the magnetization was acquired during and/or shortly after the crystallization of the nearly coeval Sobreiro and Santa Rosa Formations. In some cases, it is also carried by magnetite with high coercivities/high-blocking temperatures (samples PY96-K2, PY96-P1 in Fig. 5). The pole SF1 (319.7°E, 24.7°S, A_{95} = 16.9°) satisfies 6 out of the 7 quality criteria proposed by Van der Voo (1990) as reported below:

(1) Rhyolites of the Santa Rosa Formation are now well-dated with a U-Pb zircon age (LA-ICP-MS) of 1877.4 ± 4.3 Ma (this study), which likely corresponds to the age of magnetization.
(2) The component was calculated with 10 sites, (88 specimens) and adequate Fisher's statistical parameters (Dm = 149.2°, Im = 79°, k = 26.4, α_{95} = 9.6°, N = 10).
(3) Remanence was investigated using stepwise AF and thermal treatments, and principal component analysis (Kirschvink, 1980) was used to determine magnetic directions in orthogonal projections.
(4) A positive baked contact test performed for a Velho Guilherme dike (Fig. 5) attests to the primary nature of SF2 component. This test also serves as a reverse baked test for the SF1 component (carried by host Santa Rosa andesites), proving the SF1 is older than ~1855 Ma (the dike's age).
(5) The Paleoproterozoic units (~1880–1855 Ma) in the area of São Felix do Xingu are unmetamorphosed to anchimetamorphic (low-grade metamorphosed) and show no trace of deformation, so the Central Amazonian Province has remained stable since the collision between the Carajás and Rio Maria domains at ~2870 Ma.
(6) The SF1 pole passes the reversal test when the specimen directions are used. The presence of reversals suggests enough time has elapsed to average out secular variation.

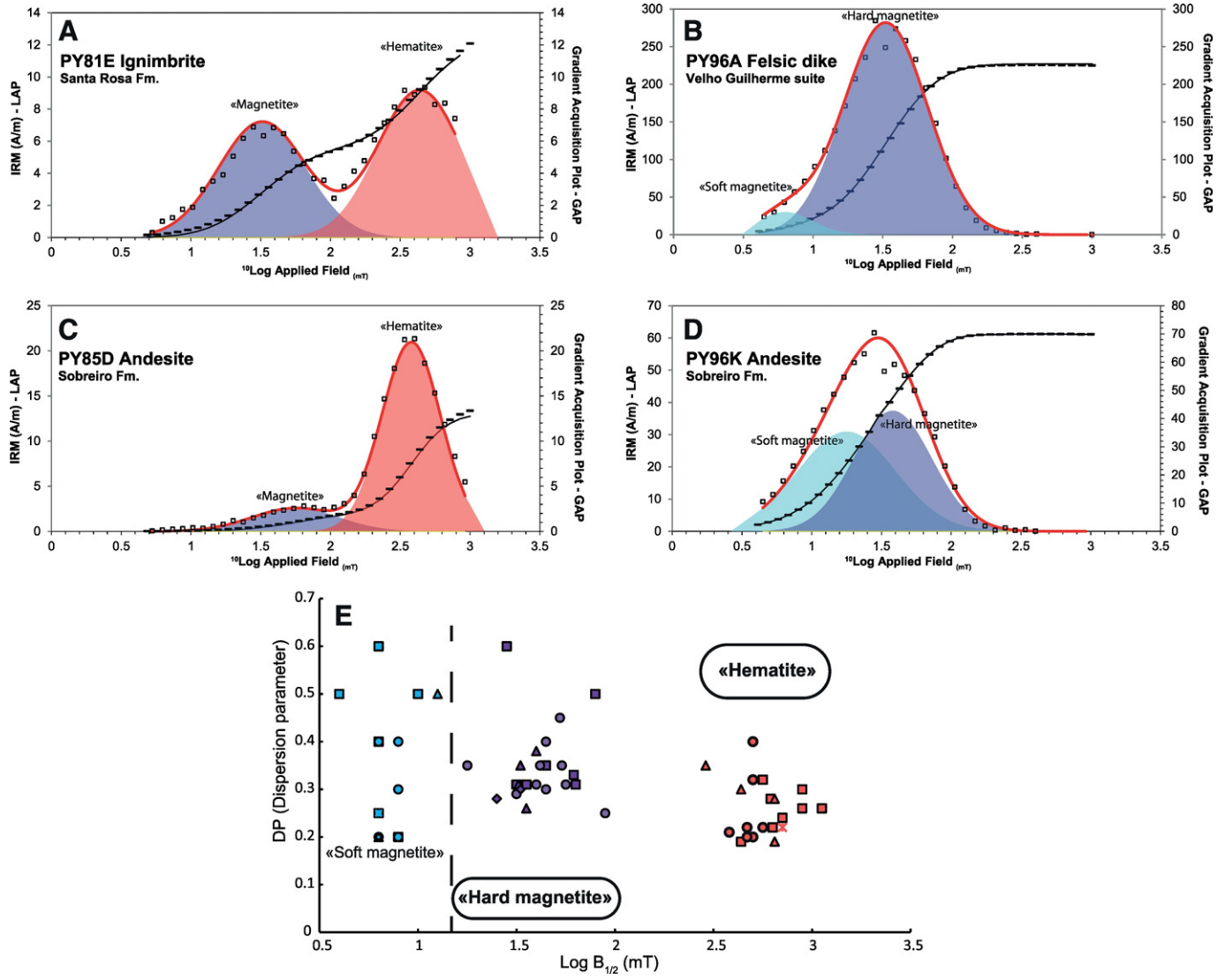


Fig. 6. Examples of characteristic isothermal remanent magnetization (IRM) after Kruiver's analysis by cumulative log-Gaussian (CLG) function, gradient acquisition plot (GAP) in red. (A) An ignimbrite showing contribution of two ferromagnetic phases with different coercivities (magnetite and hematite). (B) IRM for the felsic dike of the Velho Guilherme Suite with one mean coercivity ("hard magnetite") and a very weak component ("soft magnetite"). (C) IRM for an andesite of the Sobreiro Formation with two ferromagnetic phases (magnetite and hematite). (D) IRM for an andesite with two components of near coercivity (soft and hard magnetite). (E) The dispersion parameter in function of the mean coercivity (Log $B_{1/2}$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(7) This pole is very different from all other poles calculated for younger units in the Amazonian craton (D'Agrella-Filho et al., 2016a, 2016b) but the pole is quite close to the Early Carboniferous South American poles (Rapalini et al., 1994). There is, however, no event of this age to remagnetize these units. Moreover the steep inclination involves a high latitude position for the Amazonian craton whereas previous studies placed the craton in equatorial position at 1960 Ma with the Surumu pole (Bispo-Santos et al., 2014a) and at 1790 Ma with the Avanavero pole (Bispo-Santos et al., 2014b).

This new 1880 Ma SF1 pole can be seen as a very robust pole for the Central-Brazil Shield in the Amazonian craton.

9.2.2. The ~1855 Ma SF2 component

Component SF2 was disclosed in 110 specimens of 15 sites. This component was observed together with SF1 in rhyolites of the Santa Rosa Formation and andesites from the Sobreiro Formation at lower coercivities or unblocking temperatures, being thus carried by magnetite. The only samples where SF2 directions reach the origin in orthogonal diagram are from the Velho Guilherme Suite dike in Site 64 (PY92-96). The positive baked contact obtained at this site, and the new U-Pb (zircon) dating permit to constrain the age of SF2 component to *ca.* 1855 Ma. The mean direction for this component yielded a paleomagnetic pole (SF2) at 220.1°E, 31.1°S (A_{95} = 9.7°, K = 16.5) which may be classified as a key pole with Q = 7:

(1) The positive baked contact test attests that SF2 component was acquired at 1854 Ma (the emplacement age according to this study and consistent with a previous 1857 Ma age of Roverato (2016)).
(2) The SF2 component was calculated with 15 sites and (110 specimens) and is associated with good Fisher's statistical parameters (Dm = 239.2°, Im = 10.6°, k = 10.5, α_{95} = 12.4°, N = 15).
(3) As in the case of SF1 component, adequate demagnetization was performed, and vector diagrams and principal component analysis were employed to isolate and calculate vector directions.
(4) A positive baked contact test was obtained for the dike demonstrating the primary nature of SF2 pole.

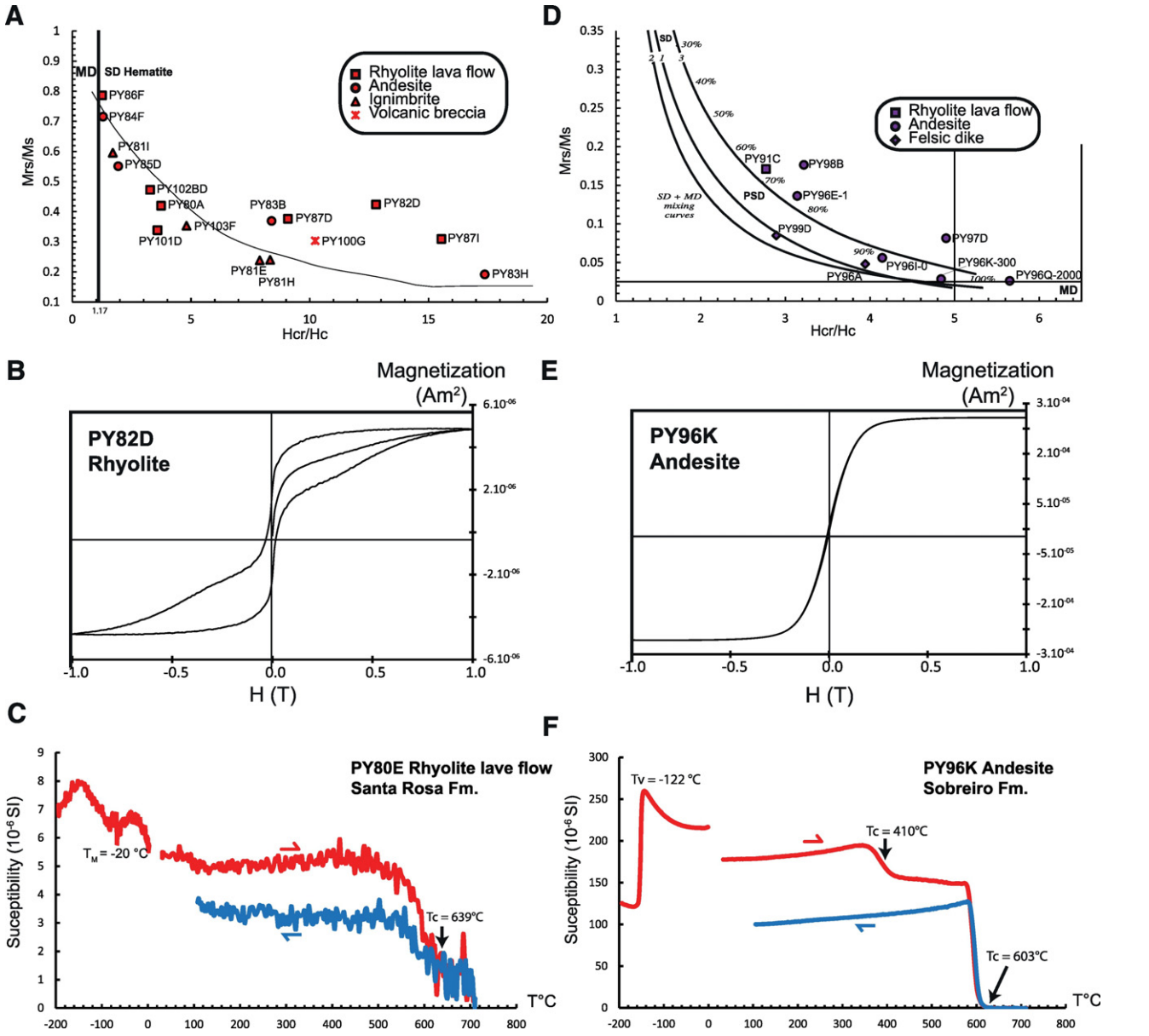


Fig. 7. (A) Day plot (Day et al., 1977) of the hysteresis ratios Mrs/Ms and Hcr/Hc for samples where the contribution of hematite is >60%. The separation between SD and MD hematite is estimated with a ratio Hcr/Hc of 1.17 according to Özdemir and Dunlop (2014). (B) Hysteresis curve for a sample of rhyolite with a strong hematite contribution (PY82D) showing wasp-waisted type (mixture phases). (C) Thermomagnetic curve with low and high temperature dependence of magnetic susceptibility showing the contribution of hematite. (D) Day plot after Dunlop (2002) for samples whose contribution of magnetite is significative, most sample fall in the Pseudo-single domain (PSD) on a typical SD + MD trend. (E) Hysteresis curve for an andesite sample (PY96K) without hematite and associated thermomagnetic curve (F).

(5) No deformation or metamorphism is observed in the dike and neither in the rhyolites and andesites in the area. This component SF2 is also present in some dikes related to the poorly dated Velho Guilherme granite located ~100 km to the east of the study area (Antonio et al., 2016).
(6) Both polarities were disclosed in the studied samples (Table 3).
(7) No similar younger paleomagnetic pole is described for the Amazonian craton.

9.2.3. Neoproterozoic to Cambrian remagnetization SF3

Component SF3 is present in 78 specimens of 14 sites from all units in the area. When it occurs together with SF1 and/or SF2, this component is the less stable, occurring at low coercivities and low unblocking temperatures. It is normally present in rocks with coarser-grained texture. IRM, hysteresis, and thermomagnetic curves show that the magnetic carrier is large multi-domain magnetite grains. The corresponding SF3 pole (128.8°E, 40.7°S, A_{95} = 15.6°) is similar to the Cambrian Puga B pole (146.9°E, 33.6°S, A_{95} = 8.4°) (Trindade et al., 2003). No younger geological event is presently known in São Felix do Xingu area that could be responsible for a remagnetization of the study rocks. A thermoviscous origin for this component is suggested and may be related to the influence of the Neoproterozoic Araguaia Belt located ~300–400 km to the east of the study area or to mafic dikes similar to those described in the Carajás domain that yielded U-Pb (on baddeleyite) ages of 535.1 ± 0.9 Ma (unpublished data, Teixeira et al., 2012; Teixeira et al., 2016). The SF3 pole (Q = 3) is thus interpreted as a remagnetization due to the Neoproterozoic Brasiliano event (~600–530 Ma) developed at the border of the Amazonian craton.

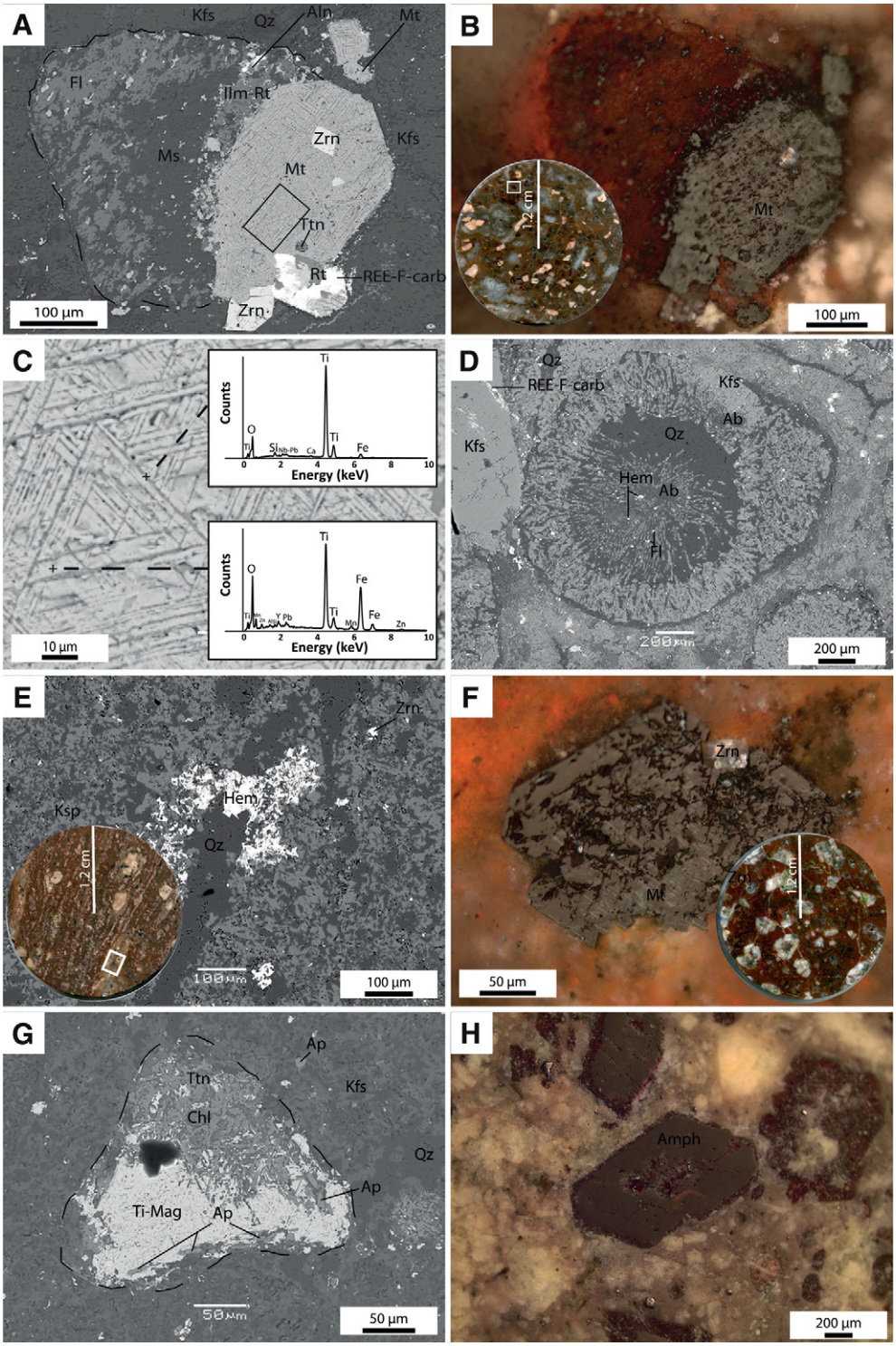


Fig. 8. Petrography of analyzed samples. (A) SEM-BSE micrograph and (B) reflected light photomicrograph (crossed nicols at 85°) of subeuhedral crystal of titanomagnetite with exsolution lamellae of ilmenite altered in hematite (Martite, Mt) (PY81D1). (C) SEM-BSE micrograph for a detailed view of the exsolutions and energy dispersive spectra EDS associated (PY81D1). (D) SEM-BSE micrograph of hematite-rich spherulite (PY81D1). (E) SEM-BSE micrograph of hematite co-precipitated with quartz veins (PY101A2). (F) Reflected light photomicrograph (crossed nicols at 85°) for a titanomagnetite crystal and (G) SEM-BSE micrograph for hydrothermal alteration in the dike of the Velho Guilherme Suite (PY92). (H) Reflected light photomicrographs (crossed nicols at 85°) of andesite (PY85C1). Mineral abbreviations used from Whitney and Evans (2010).

9.3. Paleomagnetic discrepancies between 1.9 and 1.8 Ga

With the two new poles SF1 and SF2 we can tentatively extend the APWP traced for the Amazonian craton to 1855 Ma (Bispo-Santos et al., 2014a; Théveniaut et al., 2006) (Fig. 9). In doing so, we selected the polarities that implied in the shortest APWP between 1960 and 1855 Ma. The Amazonian APWP shown in Fig. 9 defines a loop, named here the Uatumã loop and indicates a fast motion, *ca.* 39.3 cm/yr, of the Amazonian craton during the interval 1880–1855 Ma.

The paleomagnetic record of the Amazonian craton implies in rapid motions between a semi-equatorial position at 1960 Ma (Surumu pole), a polar position at 1880 Ma (SF1 pole) and a return to an equatorial

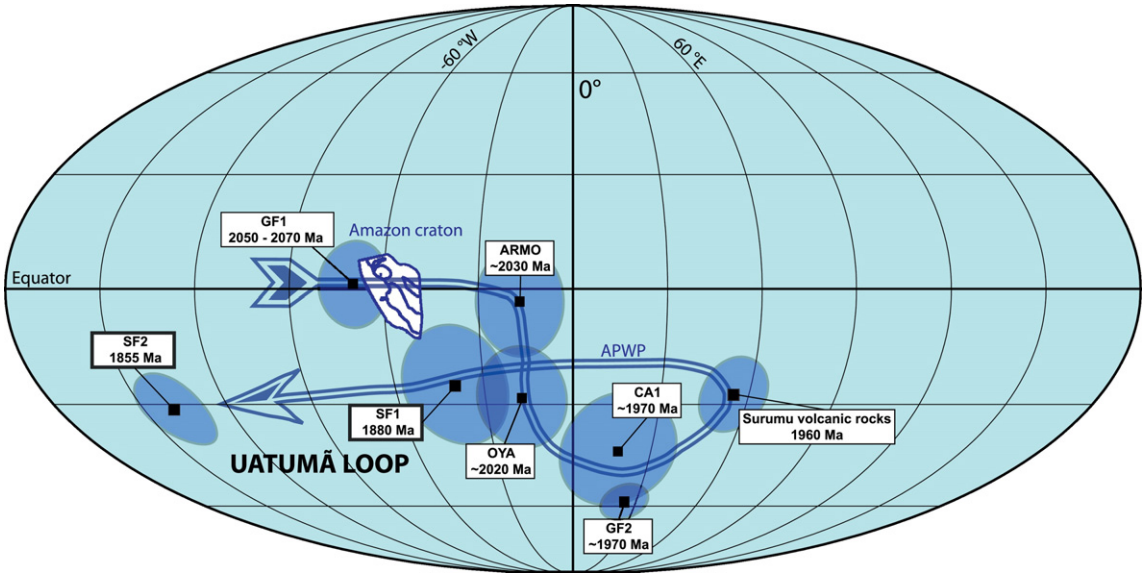


Fig. 9. APW path traced for the Amazonian craton between 2100 and 1860 Ma (paleomagnetic poles before 1880 Ma in Bispo-Santos et al. (2014a). Amazonia is in its present position.

position at 1855 Ma (SF2 pole). This behavior is similar to the paleomagnetic record of other cratons at around ~1880 Ma (Table 4).

The Slave craton in Laurentia provides the best database between 1900 and 1800 Ma. Their paleomagnetic poles (listed in the Table 4) form a ~110° loop well-known as "The Coronation loop" (McGlynn and Irving, 1978). Mitchell et al. (2010) revised data for the Slave craton and demonstrated that the loop cannot be explained parsimoniously by large or local vertical-axis rotations and interpreted the "Coronation loop" as a signal of rapid, oscillatory true polar wander (TPW). Still in Laurentia, the Superior craton provides three high quality paleomagnetic poles, all with positive magnetic stability tests. The Flaherty volcanics and Haig intrusives poles are dated at 1870 ± 1 Ma by U-Pb on baddeleyite (Hamilton et al., 2009; Schmidt, 1980) and differ by 38° from the Molson dikes B + C2 pole with an age of 1884–1877 Ma (Evans and Halls, 2010; Heaman et al., 2009). Mitchell et al. (2010) also interpret this discrepancy as a possible TPW.

Semami et al. (2016) refined the APW path of the Kaapvaal craton in South Africa during the late Paleoproterozoic (2200–1840 Ma) using a model where the Kaapvaal craton was not linked with the Zimbabwe craton. This model is based on the difference of ~38° between the Mashanaland sills pole dated at ~1883 Ma (Hanson et al., 2011; Söderlund et al., 2010) from the Zimbabwe craton and the Post-Waterberg dolerites pole dated at ~1875 Ma (Hanson et al., 2004) from the Kaapvaal craton. These authors proposed a large-scale post-1880 Ma displacement of >2000 km between the two cratons along the Limpopo Belt. The first argument against this model is the age of collision between the Zimbabwe and Kaapvaal cratons along the Limpopo Belt that occurred between 2700 and 2000 Ma (Barton et al., 2006; Nicoli et al., 2015) and no deformation is documented after 1800 Ma to accommodate a >2000 km displacement. A second argument against the model is the existence of the Sebanga-2 Virtual geomagnetic pole (Seb-2 VGP) obtained for the Zimbabwe craton (Bates and Jones, 1996). The slightly younger ~1880 Ma Sebanga dikes cross-cut the Mashanaland dolerite (~1883 Ma) but the VGP position is close to the Post-Waterberg dolerite pole (~1875 Ma) from the Kaapvaal craton (see Fig. 12 of Hanson et al. (2004)). Therefore, these results preclude any eventual displacement between Zimbabwe and the Kaapvaal cratons as proposed by Hanson et al. (2011), and the discrepancy between paleomagnetic poles is better explained by TPW.

Baltica, together with Laurentia, is the craton with the largest available paleomagnetic database. Baltica was not yet assembled at 1880 Ma and comprehend Fennoscandia (Karelia-Kola-Norbotten), Volgo-Uralia and Sarmatia cratons (Bogdanova et al., 2013). The position of Fennoscandia at 1880 Ma is well established, based on a mean paleomagnetic pole obtained from several Svecofennian gabbroid bodies (Vittangi, Kiuruvesi, Pohjanmaa-Ylivieska, Jalokoski) (Pesonen et al., 2003) and the Keuruu diabase dikes (Klein, 2016; Klein et al., 2016). The slightly younger *ca.* 1860 Ma Loftahammar gabbro pole is the only pole that can be used for comparison. A difference of ~48° is observed between the 1880 and 1860 Ma poles, but despite recent geochronological constraints Bergström et al. (2002), the Loftahammar gabbro pole cannot be considered as a key pole due to poor-quality paleomagnetic data (Poorter, 1976).

Didenko et al. (2015) revised the Paleoproterozoic APWP for the Siberia craton and four 1878–1850 Ma paleomagnetic poles with positive magnetic stability tests were selected for the time-interval of interest (Table 4). All poles have very similar geographic coordinates and therefore suggest small dislocations or rotations for Siberia in the 1880–1850 Ma interval. The ~1878 Ma Lower Akitkan pole (Malaya Fm.) is the most reliable paleomagnetic pole since it owns positive intra-formational conglomerate and fold tests. All other poles for the 1880–1850 interval were also obtained for the Akitkan volcano-

Notes to Table 4

References: 1, Buchan et al. (2016); 2, Mitchell et al. (2010); 3, Irving et al. (1972); 4, Evans and Bingham (1973); 5, Morelli et al. (2009); 6, Evans and Halls (2010); 7, Schmidt (1980); 8, Hamilton et al. (2009); 9, Williams and Schmidt (2004); 10, Findlay et al. (1995); 11, This study; 12, Hanson et al. (2011); 13, Söderlund et al. (2010); 14, Hanson et al. (2004); 15, Lubnina et al. (2010); 16, Olsson et al. (2015); 17, Didenko et al. (2009); 18, Donskaya et al. (2007); 19 Vodovozov (2010); 20, Didenko et al. (2006); 21, Larin et al. (2003); 22, Vodovozov et al. (2007); 23, Prasad et al. (1987); 24, Meert et al. (2011); 25, French et al. (2008); 26, Belica et al. (2014); 27, Radhakrishna et al. (2013b); 28, Radhakrishna et al. (2013a); 29, Elming (1985); 30, Skiöld (1988); 31, Neuvonen et al. (1981); 32, Marttila (1981); 33, Pesonen and Stigzelius (1972); 34, Helovuori (1979); 35, Mertanen and Pesonen (1992); 36, Mertanen (2013) unpublished; 37, Pesonen et al. (2003); 38, Elming (1994); 39, Klein (2016); 40, Poorter (1976); 41, Bergström et al. (2002); 42, Williams et al. (2004); 43, Rasmussen et al. (2012); 44, Idnurm and Giddings (1988).

[a] Angular separation of poles (°).

[b] Time separation of poles with the error associated (Ma).

[c] Conservative APW rate (cm/yr) with error associated (velocity maximum-velocity minimum).

Table 4
Compilation of paleomagnetic poles used in the TPW analysis. Plat, paleolatitude; Plong, paleolongitude; A_{95}, semiangle of the cone of 95% confidence about the pole. Geochronological symbols: corr. - correlation with related unit; zrn - zircon; badd - baddeleyite; pl - plagioclase; phl - phlogopite; Q - quality factors (Van der Voo, 1990). Analysis of TPW.

Name	Plat (°N)	Plong (°E)	A_{95}	Age ± error (Ma)	Q	Ref
Slave (the Coronation loop)						
Ghost dykes	2	254	6	1887 ± 5 U-Pb badd	6	1
Mean Seton/Akaitcho/Mara	−6	260	4	1885 ± 5 U-Pb corr.	5	2
Mean Kahochella/Peacock Hills	−12	285	7	1882 ± 4 U-Pb zrn corr.	4	2
Douglas Peninsular	−17	245	16	~1885–1870	4	2
Takiyuak Fm.	−13	249	8	~1885–1871	5	2
Stark Fm.	−11	199	6.7	1876 ± 10 corr.	5	2
Tochatwi formation	−14	204	15	~1876 corr.	5	2
Mean Pearson/Peninsular sills (PS)	−22	269	6	1870 ± 4 U-Pb	4	2
Et-Then Fm. (Group C)	4	310	8	~1840	6	3
Martin Fm.	−9	287	9	1818 ± 4 U-Pb badd	4	4, 5
Great Circle Calculation	*−76.9*	*56*	*17.3*			
Analysis Ghost vs Stark Fm.	***56.2° ± 13***[a]			***11 ± 15 Ma***[b]	***56.7 cm/yr (247–38)***[c]	
Superior						
Molson dykes B + C2	29	218	4	1884 ± 2 U-Pb badd	5	6
Flaherty volcanics	0	244	7	1870 ± 1 U-Pb badd	6	7, 8
Haig intrusives	1	247	6.1	1870 ± 1 U-Pb badd	6	7, 8
Sokoman Fm. A	−28.6	247	13.2	~1840–1830	4	9, 10
Sokoman Fm. B	−29.6	250.9	6.4	~1840–1831	5	9, 10
Great Circle Calculation	*23.9*	*329.1*	*14*			
Analysis Molson vs Flaherty	***38.2° ± 11***[a]			***7 ± 8 Ma***[b]	***60.6 cm/yr (78–43)***[c]	
Amazonia						
SF1	−24.7	319.7	16	1877.4 ± 4.3 U-Pb zrn	6	11
SF2	−31.1	220.1	9.7	1853.7 ± 6.2 U-Pb zrn	6	11
Great Circle Calculation	*50.3*	*263.4*	*0.2*			
Analysis SF1 vs SF2	***85° ± 25***[a]			***25 ± 10 Ma***[b]	***39.3 cm/yr (48–19)***[c]	
Kalahari		158.5				
Mashonaland sills (Zimbabwe)	6.5	338.5	5	1883 ± 2 U-Pb badd	4	12, 13
Post-Waterberg dolerites (Kaapvaal)	15.6	17.1	8.9	~1879–1872 U-Pb badd	4	14
Black Hills dyke swarm (Kaapvaal)	9.4	352	5	~1875–1835	5	15, 16
Great Circle Calculation	*−72*	*49.3*	*1.5*			
Analysis Mash vs Post-Waterberg	***38.8° ± 14***[a]			***9 ± 3 Ma***[b]	***47.9 cm/yr (49–27)***[c]	
Siberia						
Lower Akitkan, Malaya Fm.	−30.8	98.7	5	1878 ± 4 U-Pb zrn corr.	5	17
Chaya Fm. Upper Akitkan Group (Anabar) Red beds	−22.1	97.5	6.9	1863 ± 9 U-Pb zrn corr.	4	18, 19
Shumikhin granite	−23.7	110	5.7	1855 ± 5 U-Pb zrn	3	20, 21
Okun Fm.	−28.5	111	9.6	~1850	3	22, 19
Great Circle Calculation	*−63.4*	*291.3*	*32.2*			
Analysis Lower Akitkan vs Okun Fm.	***10.9° ± 15***[a]			***28 ± 9 Ma***[b]	***4.3 cm/yr (7–5)***[c]	
India						
Cuddapah Basin sediments	29.3	332.9	14.4	~1880	3	23
Bastar dolerite dykes-C (Group. 3)	29.3	331.7	15.7	1883 ± 1.5 U-Pb badd	5	24, 25
Dharwar 1.88 Ga dykes-C (overall mean 1.88–1.86 Ga)	35.9	331.1	7.7	~1883–1847 U-Pb badd	4	26
11. Mean around Cuddapah basin 1.88 Ga direction	50.5	331.4	6.4	1880	4	27
12. Overall mean India 1.97–1.88 Ga direction	49.2	332.9	4.8	~1991–1885 U-Pb badd	4	27
13. Mean around Cuddapah basin (~1860 Ma)	69.6	286.7	2.5	1847 ± 6 U-Pb badd	4	27
14. Granulite region (South - Dharmapuri)	82.5	259.1	10.3	1855 ± 9 Ar-Ar phl	5	28
15. Overall mean NNW-N shallow direction in India	73.7	282.6	2.9	1847 ± 6 U-Pb badd	4	27
Great Circle Calculation	10.6	69.1	8.7			
Analysis Bastar dolerites-Mean 1.86 Ga	**48.49° ± 18.2**[a]			**~20 Ma**[b]	**26.9 cm/yr (32–14)**[c]	
Baltica						
Vittangi gabbro	42.6	227.9	4.9	1886 ± 14 U-Pb zrn	2	29, 30
Kiuruvesi intrusions (mean)	43.1	235.2	10	1886 ± 5 U-Pb zrn	2	31, 32
Pohjanmaa-Ylivieska gabbro	38.6	239.8	10.9	1879 ± 5 U-Pb zrn	3	33, 34
Jalokoski gabbro	43.1	233.9	7.6	1871 ± 4 U-Pb zrn	2	35, 36
Mean Baltica SVF1	41	233	5	~1881	3	37
Svecofennian volcanics and intrusions	46	227	4.1	~1880	4	38
Keuruu diabase dyke swarm	45.4	230.9	5.5	1870 ± 9 U-Pb zrn	?	39
Loftahammar gabbro	23	179	4.6	1859 ± 9 U-Pb zrn	3	40, 41
Great Circle Calculation	45	64.5	15			
Analysis Mean Baltica vs Loftahammar	**48.34° ± 9.6**[a]			**~22 Ma**[b]	**24.4 cm/yr (190–4)**[c]	
Australia	Plat rot.	Plong rot.	(Li & Evans (2011) correction)			
Frere Formation red beds (WAC)	−8.26	1.8	Mean 1891 ± 8 U-Pb zrn		6	42, 43
Plum Tree volcanics (NAC)	−29	9.3	1825 ± 4 U-Pb zrn		?	44
Great circle	30.8	0.5				
Analysis FF *vs* Plum Tree	**25.2° ± 11.1**[a]			**~62 Ma**[b]	**4.3 cm/yr (8–2.3)**[c]	

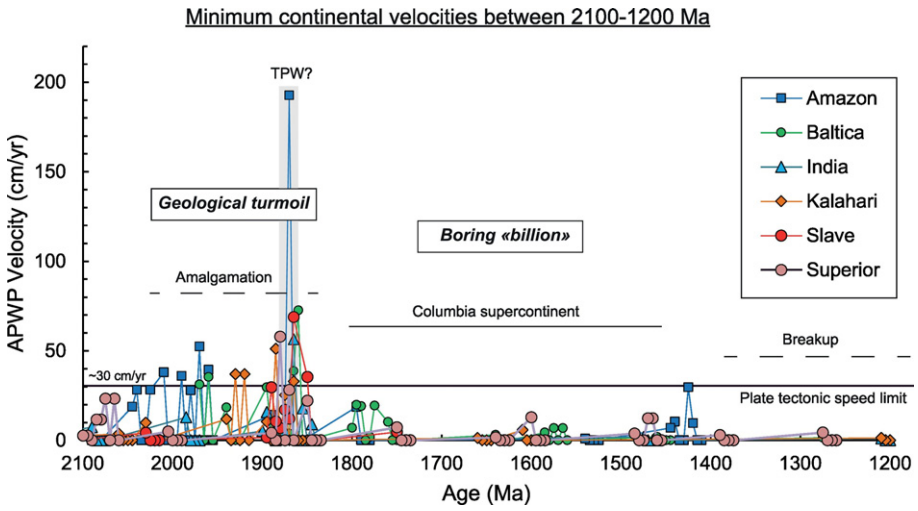


Fig. 10. Minimum continental velocities calculated between 2100 and 1200 Ma for six cratons during the Columbia supercontinental cycle. Plate tectonic speed limit is marked by a line according to the value of ~30 cm/yr (Gordon et al., 1979; Tarduno, 1990). Paleomagnetic data between 2100 and 1200 Ma is available in Table – Supplementary material.

plutonic belt, which was slightly deformed and metamorphosed (Zorin et al., 2008). The analysis of the data is thus complicated because this region has undergone remagnetizations as indicated by some negative conglomerate tests (Didenko et al., 2009).

Belica et al. (2014) performed the latest paleomagnetic study for the India craton and calculated a 1880 Ma mean pole for this block (Dharwar dikes). According to the authors, the high-quality of this mean pole is supported by a positive baked contact test and by a precise U-Pb age on baddeleyite. However, this pole represents an average between dikes of different ages and different directions. For example, site 74 is dated at 1885.4 ± 3.1 and site 19 at 1847 ± 6 Ma (U-Pb on baddeleyite). Here we will use instead the study of Radhakrishna et al. (2013b), which published two different poles for 1880 Ma and 1860 Ma for the Cuddapah Basin.

In summary, several cratons show conspicuous discrepancies between paleomagnetic poles dated between 1880 and 1860 Ma. Using the paleomagnetic database in the interval 2100–1200 Ma (Table – Supplementary material) we can estimate the Precambrian continental drift rate for six cratons with the PMTec package of Wu et al. (2015) based on the method of McFadden and McElhinny (1995). As the paleolongitude is unconstrained in paleomagnetism, we can determine only the minimum motion in paleolatitude. Increments with time windows of 20 Ma and 5 Ma were chosen to identify any large polar wander deviations between 2100 and 1200 Ma. Fig. 10 shows the APWP velocity (cm/year) for six cratons in the interval 2100–1200 Ma. The velocity increases for all cratons in the interval 1890–1850 Ma and some of them show very large velocities (>50 cm/yr). Note that velocities are much smaller after 1800–1750 Ma. Piper (2013) had already noted that during the 2000–1800 Ma period, APW paths traced for the cratons are problematic in terms of velocity and paleomagnetic results and did not exclude the possible existence of true polar wander episodes.

The first possibility to explain this period of high APWP velocity is to consider that plate motions were much faster than today. Zahirovic et al. (2015) carried an analysis of plate speeds and show that fast plate motions (>10 cm/yr) are linked to plates with large subduction zones and low continental areas. They estimated a speed limit of ~20 cm/yr for Archean/Proterozoic cratonic plates which contained <50% of present continental areas. This speed limit is at the same order of previous estimates of 18 cm/yr (Meert et al., 1993) and 30 cm/yr (Gordon, 1990; Tarduno, 1990). Rapid acceleration can be linked to anomalously hot mantle and a possible influence of plume head arrival. The most famous example is for India during the Tethyan subduction reaching a ~18 cm/yr speed (van Hinsbergen et al., 2011), which is still well slower than the suggested speeds for 1890–1850 Ma (Fig. 10).

Another possibility is to consider the absence of a stable GAD (geocentric axial dipole) field during the 1890–1850 Ma time interval. A global axial dipole field is the necessary and mandatory condition for using paleomagnetic data in paleogeographic reconstructions (Meert, 2009). Several lines of evidence suggest that the geocentric axial dipole (GAD) hypothesis is viable during the Precambrian (Evans, 1976). According to Veikkolainen et al. (2014a), new results support the existence of a GAD field in Precambrian times with a small octupolar component (5% of GAD) and a quadrupole component (2% of GAD). Using our compilation (Table – Supplementary material), we performed an inclination asymmetry test with a small number of reversals (N = 12) during the 1880–1840 Ma period. Values with strong anomalies (>20°) are not visible during this interval and would support a GAD field with low contamination of non-dipole-terms (Fig. S1). We performed also an inclination frequency analysis during the interval 2100–1200 Ma to detect a non-dipolar contamination but a small number of data (N = 122) was available in this compilation to be statistically representative (Fig. S2).

The existence of a GAD field depends primarily of the geodynamo's history that can be analyzed through the global paleointensity record. Numerical simulations over two billion years show that the geodynamo could have transitioned from a multipolar to a dipolar regime around 1700 Ma (Driscoll, 2016), with another weak dipolar period around 1000 Ma and the inner core nucleation around 650 Ma. But our knowledge about the global heat balance between core and mantle and the geodynamo are caveats for the reality on this model (Driscoll, 2016). Nonetheless, Smirnov et al. (2016) argue that paleointensity data should be used with caution to constrain inner growth or the long-term evolution of the core because of the poor quality of the database.

The last possibility involves true polar wander (TPW) events with the rotation of the whole lithosphere and mantle (on D″) with respect to the Earth's rotation axis (Raub et al., 2007). Possible signals of TPW have already been reported in the literature to explain the anomalous APWP between 1890 and 1850 Ma (Mitchell et al., 2010).

9.4. True polar wander and paleogeography at 1880–1860 Ma

Our new data are used to test whether paleogeographic reconstructions are consistent with the TPW episodes proposed by Mitchell et al. (2010). Relative motion (only in latitude) and rotation of cratons with respect to the spin axis of the earth is classically determined with paleomagnetism using the APWPs. In the TPW hypothesis, the motion of the entire lithosphere may be faster than plate tectonics, and the APWPs will reflect the amount of true polar wander (APW = TPW). The motion

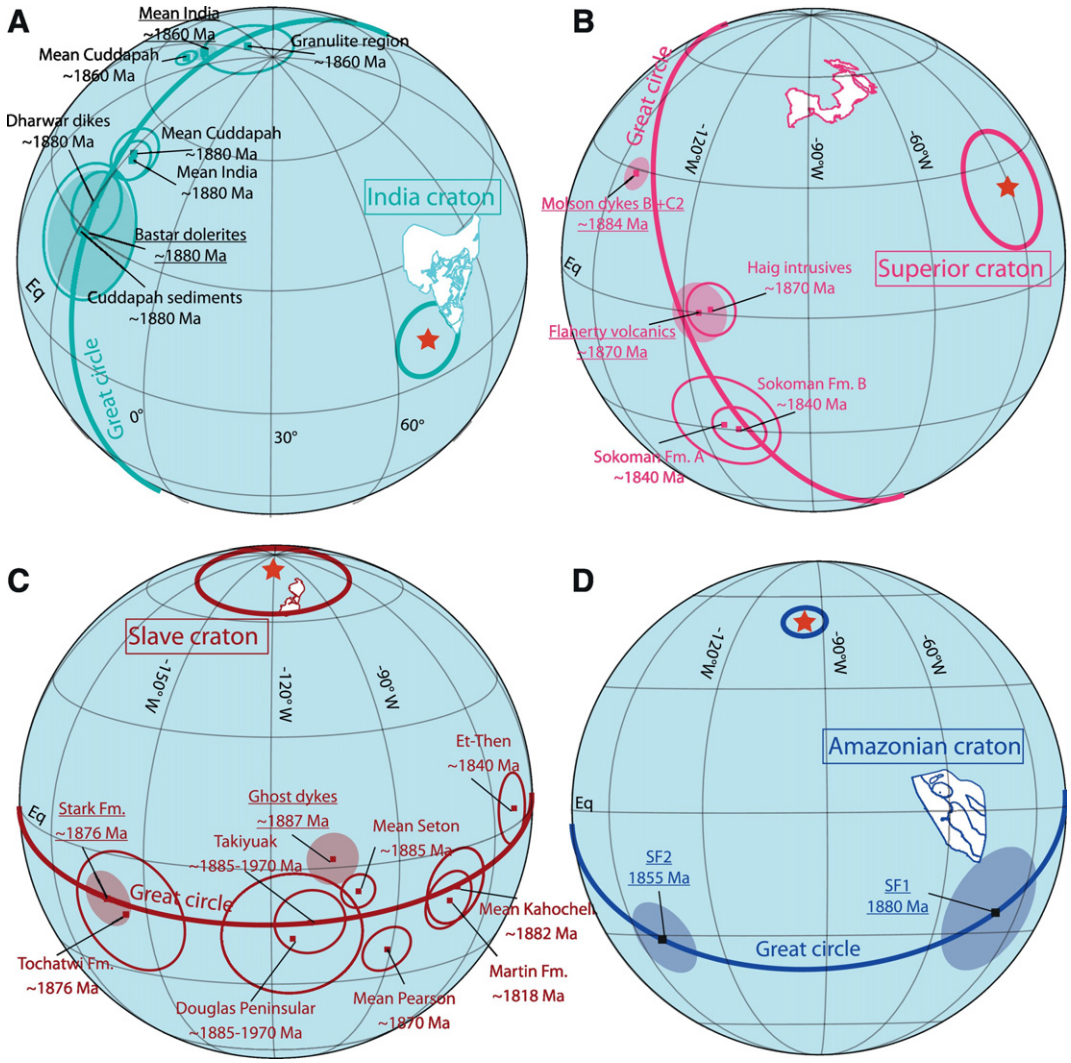


Fig. 11. Apparent polar wander paths (APWPs) length of India (A), Superior (B), Slave (C) and Amazonia (D). Paleomagnetic poles indicated are listed in Table 4 and poles used in reconstructions are underlined. Great circles calculated for each craton are indicated by thick lines and axis by red stars with their corresponding cone of confidence circles (Table 4). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of cratons is thus determined with respect to the TPW reference axis (I_{min} - minimum inertia axis) and all cratons must undergo identical TPW. The TPW reference axis is determined by the overlapping of APWPs, so the relative longitudinal position between the cratons is fixed (Kirschvink et al., 1997). Therefore, with the Precambrian database (Table 4) we can test if the APWPs from all cratons show nearly the same length and shape during the proposed TPW episode (1880–1860 Ma). We used the technique of Meert (1999) to compare APWs and evaluate the errors.

The APW path for the "Coronation loop" is the most complete and the most complex (Fig. 11) (Mitchell et al., 2010). In our paleogeographic reconstructions we used the well-dated 1887 ± 5 Ma pole of Ghost dikes with a positive baked contact test (Buchan et al., 2016) and the youngest pole of Stark Formation (~1876 ± 10 Ma) with a positive conglomerate test (Evans and Hoye, 1981; Mitchell et al., 2010). The angular distance between the two poles is 56° (±13) which implies a velocity of 56.7 $\frac{+27}{-38}$ cm/yr (interval of 11 ± 15 Ma). The calculated great circle includes all poles of Mitchell et al. (2010) with the axis (Plat = −76.9°N, Plong = 56°E, A_{95} = 17.3).

For the Amazonian craton we calculated a great circle between poles SF1 (~1880 Ma) and SF2 (~1855 Ma). With only two poles, we have necessarily a great circle and it is not possible to estimate the error (Plat = 50.3°N, Plong = 263.4°E) but the angular distance between poles of 85° ± 25 in the time interval of 24 ± 10 Ma yields a velocity of ~39.3 $\frac{+48}{-19}$ cm/yr. The greatest angular distance can be attributed to the fact that the considered time interval is longer than for the Slave craton.

A great circle was calculated for the Superior craton using three paleomagnetic poles between 1880 Ma (Molson B + C2 dikes pole) and 1870 Ma (Flaherty volcanics pole) (Plat = 40.3°N, Plong = 336°E, A_{95} = 4.1). The angular distance between the poles is 38.2° ± 11 during a short time interval of 7 ± 8 Ma (velocity of ~61 $\frac{+78}{-19}$ cm/yr). Considering the associated errors, this distance is close to that of the Slave craton.

A similar angular separation of 38.8° ± 14 is calculated for the Kalahari craton between the Mashonaland sills pole (Hanson et al., 2011) and the Post-Waterberg dolerites pole (Hanson et al., 2004). The angular distance between the ~1880 Ma mean Baltica pole (Pesonen et al., 2003) and the Loftahammar gabbro pole (1859 ± 9 Ma) (Poorter, 1976) is 48.34° ± 9.6 which is very similar to Indian angular distance (48.49° ± 18.2) between the mean 1880 Ma and 1860 Ma poles (Radhakrishna et al., 2013b). Australia has a very poor database (Schmidt, 2014). Only two poles with relatively good quality could be used, despite the much younger age of the Plum Tree volcanic pole (1825 ± 4 Ma) when compared to the poles used for other cratons.

Siberia is the only craton without any significant difference between the 1880 and 1850 Ma poles (angular distance of ~10° ± 10). So, a great circle for its poles provides a large uncertainty (A_{95} = 32.2). The fact that Siberian poles are similar across the time interval of 1880–1850 Ma may be explained either as a failure of the TPW hypothesis or as a problem with the paleomagnetic data because the sampling area was not geologically stable at ~1880–1850 Ma. In summary, most cratons have angular distances around 40–50° between 1880 Ma and 1860 Ma poles except for the Amazonian craton with ~85° and Siberia (<10°).

After calculating the best-fit great circles for each craton and rotating them to the center of a Mollweide projection (spinner projection, Plat = 0°, Plong = 0°), we can superimpose the great circle axes (Raub et al., 2007). Thus, the paleolongitude of all considered cratons are constrained. We used this hypothesized TPW axis for two reconstructions at ~1880 Ma and ~1860 Ma (Fig. 12) where the paleolatitude of any craton is constrained by the corresponding pole with two possibilities due to geomagnetic polarity ambiguity.

In the ~1880 Ma reconstruction (Fig. 12-A), all APWPs are aligned on the same great circle with a consistent distribution of poles depending on ages. The Amazonian craton is in polar position constrained by the SF1 pole (this study) and differs from the usual equatorial position in different models of supercontinent Columbia (Bispo-Santos et al., 2014b; Evans and Mitchell, 2011; Pehrsson et al., 2016; Pisarevsky et al., 2014). Siberia is situated at the center of the projection, and close to the Slave craton. Siberia is considered as the center of the future Columbia supercontinent (Evans and Mitchell, 2011; Gladkochub et al., 2016), and recent studies support a Slave-Siberia connection (Buchan et al., 2016; Ernst et al., 2016; Evans et al., 2016b). Reconstruction of Fig. 12 supports a large ocean between the Slave and the Superior cratons, the so-called Manikewan Ocean (Corrigan et al., 2009); the large distance between these cratons being already pointed by Mitchell et al. (2014) based on distinct Paleoproterozoic Slave and Superior APWPs.

In the ~1860 Ma reconstruction (Fig. 12-B) we represent a counterclockwise rotation of all cratons indicated by the arrows. The position of the Amazonian craton is equatorial and close to Baltica (Fennoscandia) consistent with a SAMBA-type connection (see D'Agrella-Filho et al., 2016a, 2016b for a review). After rotation, the Siberia-Slave connection is possible and continental blocks appear to converge toward an assembled supercontinental configuration. The collision between Slave and

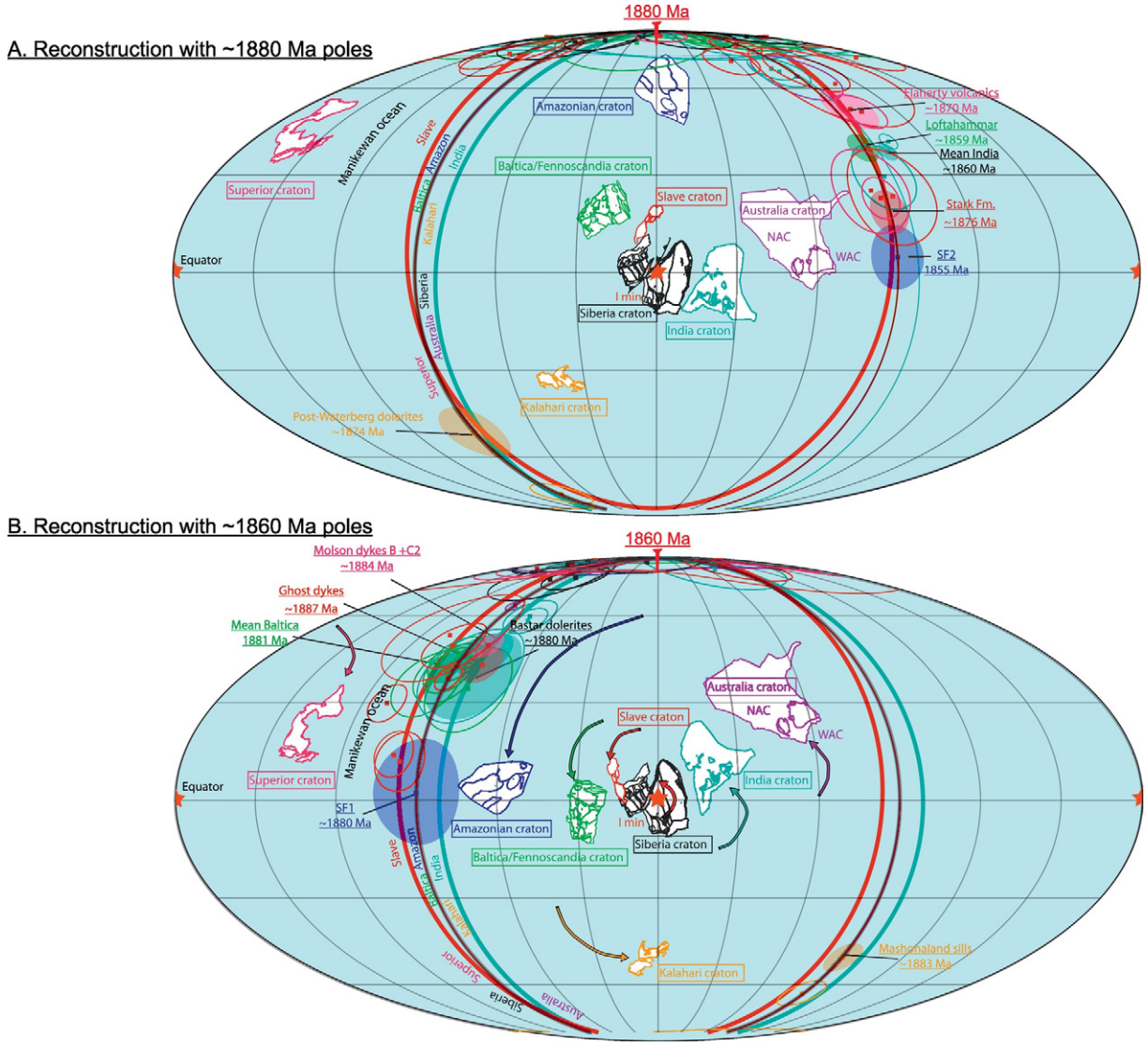


Fig. 12. Possible paleogeographic reconstructions at 1880 Ma (A) and 1860 Ma (B) during the proposed true polar wander. The reconstructions are established with the true polar wander spin axis (I_{min}) as a red star in the center (spinner diagram) and the great circles associated at each craton with the repartition of paleomagnetic poles. Cone of confidence circles for the error of great circle axis are not illustrated for visibility. Paleomagnetic poles used for the reconstructions are labeled with semitransparent filled ellipse. The sense of motion is indicated by a red arrow. *Abbreviations*: NAC = North Australian craton, WAC = West Australian craton. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Superior occurred during the development of the Trans-Hudson orogeny (1860–1830 Ma). The final assembly of Laurentia would be at *ca.* 1715 Ma after the collision of Wyoming craton with the Superior craton. This reconstruction does not support a connection between Baltica and India as supposed by Pisarevsky et al. (2014). The longitudinal position of Australia craton is consistent with a future connection with Laurentia at ~1650–1600 Ma along the Racklan orogeny (Thorkelson and Laughton, 2015). Finally, Kalahari is considered in this model as an isolated block without evidence of connection with any other craton. It is interesting to note that in our model, where paleolongitude (with respect to TPW) is constrained, positions of the different cratons are not so different from those in reconstructions based on individual paleomagnetic poles alone (Pehrsson et al., 2016; Zhang et al., 2012).

Despite the small amount of available data, our results support a TPW event between 1880 Ma and 1860 Ma with the rotation of all continental blocks around a minimum inertia axis located on the paleoequator, through Siberia (I_{min} – Fig. 12).

9.5. Geological turmoil during the amalgamation of the supercontinent Columbia

U-Pb ages of Precambrian detrital zircons define major peaks at 2700, 2500, 2100, 1880 and 1100–1000 Ma (Fig. 13-A) (Belousova et al., 2010; Condie and Aster, 2010). These peaks immediately follow times of enhanced mantle activity and witness steps of increased crustal growth. Moreover, Hf isotope compositions in zircons help to determine their provenance, either from juvenile (mantle-derived) material or from recycled continental crust (Fig. 13-B) (Kemp et al., 2006). Whereas a juvenile origin predominates at 2700 and 2500 Ma, a recycled source predominates at 2100 and 1880 Ma (Arndt and Davaille, 2013). We regard this observation as evidence that amalgamation of the Columbia supercontinent was already ongoing during the time interval of interest (1880–1860 Ma). Indeed, the time interval between 2100 and 1800 Ma corresponds to several collisional/accretional orogenic belts in many Precambrian terranes worldwide (Condie, 2002; Hoffman, 1988; Zhao et al., 2002).

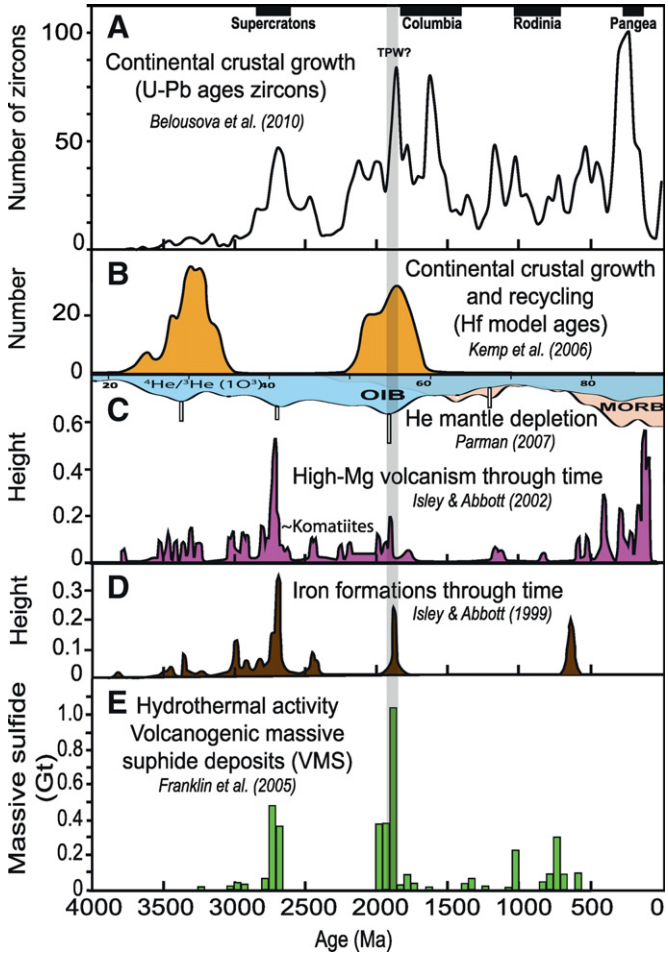


Fig. 13. Geological features during the Precambrian record between 4000 and 500 Ma. (A) Histogram for the crustal zircon age distribution of Belousova et al. (2010). (B) Peaks in Hf model ages in orange show times of juvenile crustal addition (Kemp et al., 2006). (C) $^4He/^3He$ probability density function from Parman (2007) with peaks of OIB (ocean island basalt) and MORB (mid-ocean ridge basalt) correlated to the frequency of the High-Mg melts though time (Isley and Abbott, 2002). (D) Frequency of iron formations (BIFs) through time (Isley and Abbott, 1999). (E) Distribution of the volcanogenic massive sulphide deposits (VMS) from Franklin et al. (2005).

Fig. 13-C also shows the frequency of mantle-derived melts through time (Isley and Abbott, 2002). The most important peak in high-MgO magmatism is at 2700 Ma, and is well-known for its abundant komatiites. The 1880 Ma peak also corresponds to a peak in the komatiite record, being the last ultramafic volcanic pulse of Precambrian times. The global LIP record on all cratons is available in (Condie et al., 2015; Ernst, 2014). For the period of interest, we can cite the 1880 Ma NE-trending Ghost dike swarm in the Slave craton, the 1880 Ma Circum-Superior LIP (Molson dikes and Flaherty volcanics) in the Superior craton (Minifie et al., 2013), the 1880 Ma Southern Bastar-Cuddapah LIP in India (French et al., 2008), the Mashonaland sills and the Post-Waterberg dolerites in Kalahari craton (Hanson et al., 2004), Svecofennian A-type magmatism in Baltica and in Siberia, and the Uatumã LIP in Amazonia craton (this study). Fig. 13-E shows a major peak in the formation of volcanogenic massive sulphide (VMS) deposits related to a strong hydrothermal activity around 1880 Ma also suggesting an important production of mantle-derived magmas. This period also coincides with a major peak in orogenic gold resources (Goldfarb et al., 2001) and iron formations (BIFs and GIFs) (Fig. 13-D) (Bekker et al., 2010; Isley and Abbott, 1999). So, understanding the geodynamic of this period has an economic interest.

In summary, the Precambrian geological record suggests a high production of mantle-derived magmas at around 1880 Ma, together with the beginning of the Columbia amalgamation and a peak in the magmatic recycling of continents. Both mantle superswells and thermal insulation under an amalgamating supercontinent are causes of density perturbations that alter the Earth's inertia tensor, potentially provoking TPW episodes. These events will place the supercontinent and/or the superswells at the Equator. These conditions could be linked to a whole mantle reorganization following magmatic shutdown between 2400 and 2200 Ma (Arndt and Davaille, 2013; Condie et al., 2009).

10. Conclusions

The following conclusions can be drawn from this work:

(1) Two new paleomagnetic poles considered as primary are found for the Amazonian craton. The SF1 pole from rhyolites is dated at 1877.4 ± 4.3 Ma (U-Pb zrn, LA-ICP-MS) and its primary origin is supported by an inverse contact test, reversal polarities and the late-to post-magmatic hydrothermal origin of hematite as its carrier. The SF2 pole is a secondary component in the ~1880 Ma rhyolites and a primary component in the dike of Velho Guilherme Suite dated at 1853.7 ± 6.2 Ma (U-Pb zrn, LA-ICP-MS). Its primary origin is supported by a positive baked contact test and implies a remagnetization of magnetite at ~1855 Ma in the rhyolites.

(2) The SF3 pole is considered as a widespread remagnetization event in the area affecting the low-coercivity and low-blocking temperature phases in all units of the region. The SF3 component is interpreted to be related to a Neoproterozoic remagnetization during the Araguaia Belt development.

(3) The significant discrepancy of Paleoproterozoic poles between 1880 and 1860 Ma could be explained by a true polar wander

(TPW) event which is supported by paleomagnetic reconstructions at 1880 Ma and 1860 Ma and geological/geochronological evidence. This is a likely consequence of the reorganization of mantle convection. If this model is confirmed, some reevaluation will be required about the paleomagnetic approach and the geology and paleogeography of the Columbia supercontinent.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.gr.2017.05.006.

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